Side-Channel Resilience Lattice-Based Post-Quantum Cryptography

Submitted in partial fulfillment of the requirements of the degree of

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in
Artificial Intelligence and Machine Learning

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Abstract

As quantum computing technology continues to evolve, it poses a significant threat to classical cryptographic systems by enabling efficient solutions to problems once considered intractable. Quantum algorithms like Shor's algorithm can break widely-used encryption schemes such as RSA, ECC, and ECDSA by exploiting the principles of quantum superposition and entanglement. In response, the cryptographic community has turned to lattice-based Post-Quantum Cryptography (PQC) as a promising alternative due to its strong security assumptions and resistance to quantum attacks. However, while lattice-based schemes are theoretically secure against quantum adversaries, their practical implementations remain vulnerable to side-channel attacks (SCAs) including power, timing, and electromagnetic (EM) analysis which can leak sensitive information during cryptographic operations. A common source of leakage is the discrete Gaussian sampling process, often implemented using the Centered Binomial Distribution (CBD) method, which is not inherently constant-time. This work proposes the integration of a constant-time Knuth-Yao sampling algorithm as a replacement for the CBD sampler in NIST-standardized lattice-based schemes. By leveraging Knuth-Yao's tree-based approach with time-independent traversal, our method significantly mitigates side-channel vulnerabilities. This enhances the overall robustness of lattice-based PQC implementations against timing, EM, and power-based attacks—offering a secure and efficient foundation for post-quantum encryption in real-world applications

***Keywords**:* Post-Quantum Cryptography (PQC), Lattice-based Cryptography, Side-Channel Attacks (SCAs), Constant-Time Implementation, Discrete Gaussian Sampling, Knuth-YaoSampling (or Knuth-Yao Algorithm), Centered Binomial Distribution (CBD), Timing Attacks, Power Analysis, Implementation Security, NIST PQC Standardization.

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List of Abbreviations

Sr. No.	Abbreviation	Expanded form
1	**PQC**	Post-Quantum Cryptography
2	**NIST**	National Institute of Standards and Technology
3	**SCA**	Side-Channel Attack
4	**LWE**	Learning With Errors
5	**CBD**	Centered Binomial Distribution
6	**KY**	Knuth-Yao
7	**KEM**	Key Encapsulation Mechanism
8	**PRNG**	Pseudo-Random Number Generator
9	**NTT**	Number Theoretic Transform
10	**FFT**	Fast Fourier Transform
11	**RSA**	Rivest-Shamir-Adleman
12	**ECC**	Elliptic Curve Cryptography
13	**ECDSA**	Elliptic Curve Digital Signature Algorithm
14	**EM**	Electromagnetic
15	**DPA**	Differential Power Analysis
16	**SPA**	Simple Power Analysis

17	**DDG**	Discrete Distribution Generating (Tree)
18	**QPRF**	Quantum Pseudo-Random Function
19	**MLWE**	Module Learning With Errors
20	**SIS**	Short Integer Solution
21	**SIVP**	Shortest Independent Vector Problem
22	**FPGA**	Field-Programmable Gate Array

Chapter 1
Introduction

The concept of post-quantum cryptography (PQC) emerged as a response to the anticipated capabilities of quantum computers. Unlike quantum cryptography, which uses quantum mechanics directly in communication systems, post-quantum cryptography seeks to develop algorithms that run on classical computers but are secure against quantum attacks. PQC algorithms rely on hard mathematical problems that remain difficult even for quantum computers, such as lattice-based, hash-based, code-based, and multivariate polynomial cryptographic systems.

The urgency of this research is underscored by ongoing efforts from organizations such as the National Institute of Standards and Technology (NIST), which initiated a standardization process to identify and approve quantum-resistant algorithms. These algorithms aim to protect the integrity of data well into the future, even as quantum technology matures. However, despite their theoretical security, practical implementations of lattice-based PQC algorithms are susceptible to side-channel attacks (SCAs). These attacks exploit physical leakages—such as variations in execution time, power consumption, or electromagnetic emissions—to extract secret information. One of the most vulnerable components in these schemes is the discrete Gaussian sampling process, often realized through the Centered Binomial Distribution (CBD). While computationally efficient, CBD-based sampling is not inherently constant-time and may leak information based on input-dependent timing behavior.

To strengthen the side-channel resilience of lattice-based schemes, this paper proposes a constant- time implementation of Knuth-Yao sampling, a tree-based method that avoids data-dependent control flow. Applied to CRYSTALS Kyber, the approach reduces timing leakage with minimal performance overhead, offering a more secure foundation for real-world PQC deployments. This report is systematically structured into eight comprehensive sections, as outlined in the Table of Contents, to ensure clarity and logical flow of information. It commences with a detailed introduction that provides the background and context necessary to understand the problem domain, along with the motivation behind undertaking this work. This is followed by an extensive literature survey that highlights existing research, methodologies, and technologies relevant to the topic, and helps establish the foundation for identifying research gaps.

Subsequently, the report presents a well-defined problem statement along with clearly stated objectives that guide the direction of the proposed solution. The core sections delve into the design and development of the proposed methodology, showcasing its novelty and implementation details. This is further supported by a thorough analysis of the results obtained from experiments or practical implementation.

1.1 Motivation

The motivation to work on post-quantum cryptography (PQC) lies in its critical role in securing digital systems against future quantum computing threats, which will render traditional cryptographic schemes like RSA and ECC obsolete. PQC algorithms, including Kyber, Dilithium, Falcon, and SPHINCS+, are designed to resist quantum and classical attacks, making them essential for safeguarding sensitive information in sectors like banking, defence, and healthcare. However, practical vulnerabilities like sidechannel attacks and challenges in implementing these algorithms on resource-constrained devices remain significant barriers to their deployment. Addressing these gaps not only enhances global cybersecurity but also aligns with NIST's standardisation efforts, ensuring the algorithms' readiness for widespread adoption. This emerging field offers abundant opportunities for innovative research and career growth, making it a rewarding and impactful area to explore.

1.2 Problem Statement

The field of cryptography is on the precipice of a foundational shift, driven by the rapid advancement of quantum computing. The development of large-scale, fault-tolerant quantum computers poses a direct and imminent threat to global cybersecurity. Classical cryptographic systems that underpin modern digital security including RSA, ECC, and ECDSA base their security on mathematical problems like integer factorization and discrete logarithms. These problems, while intractable for classical computers, can be solved efficiently by quantum systems running algorithms like Shor's algorithm, rendering our current security infrastructure obsolete.

In response, the global cryptographic community, led by initiatives like the NIST Post-Quantum Cryptography (PQC) standardization process, has identified and promoted new, quantum-resistant algorithms. Among the most promising candidates are lattice-based cryptographic schemes, such as CRYSTALS-Kyber, CRYSTALS-Dilithium, and FALCON, which are valued for their strong security assumptions and computational efficiency.
However, a critical and dangerous gap exists between the *theoretical* security of these algorithms and their practical implementation. While these PQC schemes are mathematically secure against quantum adversaries, their physical implementations on real-world hardware remain highly vulnerable to a different class of attacks: classical side-channel attacks (SCAs).
Side-channel attacks do not target the underlying mathematical hardness of the algorithm. Instead, they exploit physical information leaked by the hardware during its operation. By meticulously observing physical leakages such as variations in execution time, power consumption, or electromagnetic (EM) emissions an attacker can reverse-engineer intermediate values and extract sensitive information, including the secret keys that the entire system is designed to protect.

Therefore, the central problem is that while NIST has approved these algorithms for their quantum resistance, they do not inherently protect against side-channel attacks, and these implementation level flaws continue to expose them to practical exploitation. This project directly addresses this vulnerability, focusing on eliminating side-channel leakage at its source. It aims to bridge the critical gap between theoretical post-quantum security and the robust, hardened, real-world implementations necessary to secure our digital future.

1.3 Objectives

- Analyze the operational design of Kyber, Dilithium, FALCON, and SPHINCS+ to identify theoretical vulnerabilities to classical side-channel attacks.

- Investigate non-constant-time operations, specifically the Centered Binomial Distribution (CBD) sampler, as a primary source of timing and power leakage in lattice-based schemes.

- Compare the execution time variations and leakage profiles of the standard implementations of these algorithms to establish a baseline for vulnerability .

- Research existing countermeasures, such as masking and noise injection, and identify constant-time execution as the most robust mitigation strategy for timing attacks.

- Propose the integration of a constant-time Knuth-Yao (KY) sampling algorithm as a secure, drop-in replacement for the vulnerable CBD sampler.

- Implement the modified KY sampler within the reference codebases of CRYSTALS-Kyber, Dilithium, and FALCON to create hardened, side-channel-resistant prototypes.

- Conduct a rigorous comparative timing analysis to empirically measure and validate the "constant-time" property of the new KY implementations versus the variable-time CBD originals.

- Verify the functional correctness of the modified algorithms to ensure that the security enhancements do not alter the cryptographic output or protocol behavior.

- Provide actionable recommendations for developers to bridge the gap between theoretical PQC security and the practical, robust deployment of side-channel-resistant algorithms in real-world applications.

1.4 Scope

The scope of this project is precisely defined to ensure a thorough analysis of a specific, critical vulnerability within the domain of Post-Quantum Cryptography (PQC). The project is centered on the practical implementation-level security of PQC algorithms rather than their underlying mathematical foundations.

This research includes a detailed comparative analysis of the side-channel attack surfaces of four NIST-standardized PQC algorithms: CRYSTALS-Kyber, CRYSTALS-Dilithium, FALCON, and SPHINCS+. The primary focus is on identifying vulnerabilities related to classical side-channel attacks, with a specific emphasis on timing attacks that exploit execution time variations. The investigation pinpoints the discrete Gaussian sampling process—specifically the Centered Binomial Distribution (CBD) sampler as a significant source of timing leakage in the lattice-based schemes (Kyber, Dilithium, and FALCON).

To address this vulnerability, the project's scope involves the research, implementation, and evaluation of a single, targeted countermeasure: the integration of a constant-time Knuth-Yao (KY) sampling algorithm. This solution is implemented by modifying the reference C codebases of the lattice-based algorithms. The evaluation is conducted within a simulated environment, using statistical analysis (such as Welch's t-test) to rigorously validate the constant-time properties of the proposed solution and compare its performance against the original CBD sampler . The functional correctness of the modified cryptographic schemes is also verified to ensure the replacement does not alter the protocol's output .

The project's scope is explicitly limited and does not include the physical execution of power (DPA) or electromagnetic (EM) side-channel attacks on target hardware. The analysis relies on statistical timing data from a simulated software environment. Furthermore, this work does not attempt to break the underlying mathematical hardness (e.g., Learning With Errors) of the PQC algorithms. While other countermeasures like masking or noise injection are identified, their implementation is outside the scope of this project, which focuses exclusively on achieving side-channel resilience through constant-time execution of the sampling module.

Chapter 2

Review of Literature

The literature review provides a comprehensive understanding of the existing research and developments at the critical intersection of Post-Quantum Cryptography (PQC) and implementation-level security. It highlights key contributions from standardization bodies, analyzes foundational PQC algorithms, and surveys the critical findings that have shaped the current knowledge of their practical vulnerabilities. By analyzing these previous works, this section establishes the significant gap between theoretical quantum resistance and practical side-channel resilience. This review serves as the foundation upon which the present study's problem statement and proposed solution are built.

The National Institute of Standards and Technology (NIST) initiated a Post-Quantum Cryptography (PQC) standardization project to evaluate quantum-safe public key cryptosystems. Among the leading algorithms selected for standardization are CRYSTALS-Kyber for Key Encapsulation Mechanism (KEM) and CRYSTALS-Dilithium and FALCON for digital signatures. However, like their classical counterparts, these PQC schemes are vulnerable to side-channel attacks (SCAs).

Table I: Literature Survey

Sr. No.	Title & Author(s)	Year	Published In	Key Findings
1	Title: Side-channel and Fault-injection attacks over Lattice-based Post-quantum Schemes (Kyber, Dilithium): Survey and New Results Author(s): P. Ravi et al.	2024	ACM Trans. Embed. Comput. Syst.	Provided a comprehensive survey of fault-injection and side-channel vulnerabilities in Kyber and Dilithium and introduced new results on their practical exploitability.
2	Title: When NTT Meets SIS: Efficient Side-Channel Attacks on Dilithium and Kyber Author(s): Z. Qiao et al.	2023	Cryptology ePrint Archive	Performed an efficient SCA by targeting the Number Theoretic Transform (NTT) implementation in Kyber and Dilithium, which is core to lattice-based constructions.
3	Title: Breaking FALCON Post-Quantum Signature Scheme through Side-Channel Attacks Author(s): M. Aysu et al.	2022	Proc. 59th ACM/IEEE Design Automation Conf. (DAC)	Exposed Falcon to SCAs by demonstrating how its Fast Fourier Transform (FFT)-based lattice arithmetic leaks critical information.
4	Title:Single-Trace Side-Channel Attacks on CRYSTALS-Dilithium Author(s): Y. Liu et al.	2023	Cryptology ePrint Archive	Showed that single-trace attacks on Dilithium are feasible by using power analysis to target polynomial multiplications.
5	Title: Profiling Side-Channel Attacks on Dilithium: A Small Bit-Fiddling Leak Breaks It All Author(s): H. Steffen et al.	2023	Proc. 14th Int. Workshop Cryptography (PQCrypto)	Conducted profiling attacks on Dilithium, showing that a small bit-leak in NTT operations can compromise the private key.

6	Title: Exploiting Small-Norm Polynomial Multiplication with Physical Attacks: Application to CRYSTALS-Dilithium Author(s): O. Bronchain et al.	2023	Cryptology ePrint Archive	Exploited small-norm polynomial multiplications in Dilithium to mount physical attacks.
7	Title: Machine Learning Based Blind Side-Channel Attacks on PQC-Based KEMS: A Case Study of Kyber KEM Author(s): P. Ravi et al.	2023	Proc. IEEE/ACM Int. Conf. Comput.-Aided Design (ICCAD)	Presented machine learning-based blind SCAs on Kyber KEM, demonstrating that even low-order leakages can be magnified using advanced AI models.
8	Title: Magnifying Side-Channel Leakage of Lattice-Based Cryptosystems with Chosen Ciphertexts: The Case Study of Kyber Author(s): Z. Xu et al.	2020	Cryptology ePrint Archive	Showed that chosen ciphertexts can amplify leakage, which, when combined with ML-based profiling, yields successful key recovery.
9	Title: More Practical Single-Trace Attacks on the Number Theoretic Transform Author(s): P. Pessl and R. Primas	2019	Cryptology ePrint Archive	Showed the feasibility of single-trace attacks on NTT in lattice cryptosystems using template-based profiling.
10	Title: Carry Your Fault: A Fault Propagation Attack on Side-Channel Protected LWE-based KEM Author(s): S. Kundu et al.	2023	Cryptology ePrint Archive	Proposed fault propagation techniques for LWE-based KEMs like Kyber, showing that even hardened implementations can leak under fault attacks.
11	Title: Cache-timing Attack Detection and Prevention Application to Crypto Libs and PQC Author(s): S. Carre et al.	2018	Proc. 10th Int. Workshop COSADE	Explored cache-timing attacks and their detection, applying their techniques to crypto libraries including PQC schemes.

12	Title: Investigating CRYSTALS-Kyber Vulnerabilities: Attack Analysis and Mitigation Author(s): A. Kundu et al.	2024	IEEE	Investigated countermeasures for Kyber but revealed their inefficacy against side-channel leakage vectors.
13	Title: Defeating low-cost countermeasures against side-channel attacks in lattice-based encryption: A case study on Crystals-Kyber Author(s): P. Ravi et al.	2023	Cryptology ePrint Archive	Demonstrated how low-cost countermeasures fail under advanced SCAs, emphasizing the need for robust protection.
14	Title: The learning with errors problem (Invited Survey) Author(s): O. Regev	2021	IEEE	Foundational surveys on LWE and lattice problems that contextualize the cryptographic hardness of PQC schemes like FALCON.
15	Title: On lattices, learning with errors, random linear codes, and cryptography Author(s): O. Regev	2024	Arxiv	Provides a foundational survey describing the Learning with Errors (LWE) problem and its relationship to lattice problems, which together form the theoretical security basis for lattice-based PQC schemes like FALCON.
16	Title: Pre-Quantum to Post-Quantum Cryptography Transition: Integration of AI, Quantum Computing, and Semiconductor Technology Author(s): S. Das and A. Das	2024	IEEE	Explored the intersection of quantum computing, AI, and cryptographic transitions, suggesting that as PQC becomes mainstream, integrated threat vectors will evolve.

17	Title: RSA Key Recovery from Digit Equivalence Information Author(s): C. Chuengsatiansup et al.	2021	Cryptology ePrint Archive	Showed that even classical cryptographic implementations like RSA can be compromised through fine-grained digit equivalence leakages, reinforcing the threat model for PQC.

2.1 Side-channel and Fault-injection attacks over Lattice-based Post-quantum Schemes (Kyber, Dilithium): Survey and New Results [1]

The ongoing standardization and imminent global deployment of Post-Quantum Cryptography (PQC), spearheaded by the NIST standardization process, have brought the practical security of these new algorithms into sharp focus. While the underlying mathematical problems of lattice-based schemes like CRYSTALS-Kyber (the selected standard for Key Encapsulation Mechanisms, KEM) and CRYSTALS-Dilithium (the selected standard for digital signatures) are believed to be secure against quantum-based attacks, their physical implementations on real-world devices remain vulnerable to a class of powerful physical attacks. This paper by Ravi et al. presents a systematic and dual-pronged contribution to this critical area. It first provides a comprehensive survey and classification of known Side-Channel Attacks (SCA) and Fault-Injection Attacks (FIA) against these two schemes. It then introduces and validates novel, customized countermeasures designed to offer simultaneous protection against this diverse threat landscape, addressing a significant gap between theoretical security and practical, hardened implementation.

The survey portion of the work meticulously organizes the existing body of research on physical attacks. The authors analyze a wide variety of reported vulnerabilities, classifying them to understand their underlying principles, similarities, and differences. This classification reveals that attacks are not monolithic; they target various components of the cryptographic implementations. For instance, in Kyber, known SCA vectors include Correlation Power

Analysis (CPA) targeting the Number Theoretic Transform (NTT), a core component of polynomial multiplication. Other single-trace attacks have been successfully demonstrated against modules responsible for Keccak hashing, message encoding, and decoding. For Dilithium, the survey notes similar vulnerabilities in its polynomial multiplication routines and highlights how fault attacks can compromise its deterministic signing variant. This systematic categorization underscores a critical challenge: the sheer diversity of attack vectors makes it exceptionally difficult for generic, one-size-fits-all countermeasures (such as masking or shuffling) to provide simultaneous and complete protection, especially on resource-constrained embedded systems where such defenses can incur prohibitive performance overhead.

Building on this comprehensive survey, the paper's primary contribution is the proposal of new, customized countermeasures specifically tailored for Kyber and Dilithium. The authors argue that a more effective approach is to integrate several distinct SCA and FIA protections within a single, unified, and optimized design. Among these new contributions are novel defensive mechanisms specifically engineered to mitigate a range of side-channel and fault-based chosen-ciphertext attacks (CCA) against the Kyber KEM, a particularly potent attack class. The goal of these custom countermeasures is to serve either as efficient standalone solutions or as robust reinforcements to existing generic defenses like masking, providing a layered security architecture.

To validate the real-world feasibility and practicality of their proposed countermeasures, the authors conducted a thorough experimental evaluation. The new defensive designs were implemented within two prominent and widely-used public PQC software libraries: pqm4, which targets the ARM Cortex-M4 microcontroller (a common platform for embedded systems), and liboqs (Open Quantum Safe), which was tested on a Raspberry Pi 3 Model B Plus featuring an ARM Cortex-A53 processor (representing a more powerful class of embedded device). This two-pronged evaluation provides a realistic assessment of the solutions' applicability. The paper's performance evaluation reveals that these integrated, custom countermeasures incur reasonable and acceptable performance overheads on both target platforms.

In conclusion, this paper makes a significant and timely contribution to the field of PQC implementation security. By systematically surveying the vast landscape of physical attacks on the NIST standards Kyber and Dilithium, it highlights the inadequacy of relying solely on generic countermeasures. The work then convincingly argues for the necessity and feasibility of custom, algorithm-specific defensive measures. By designing, implementing, and validating novel countermeasures that provide simultaneous protection against multiple SCA and FIA threats with manageable overhead, the authors provide a tangible pathway for developers to harden PQC implementations against the realistic and potent threat of physical attacks, ensuring that the next generation of cryptography is secure not only in theory but also in practice.

2.2 When NTT Meets SIS: Efficient Side-Channel Attacks on Dilithium and Kyber [2]

This paper by Z. Qiao et al. introduces a novel and highly efficient hybrid attack methodology against the NIST-selected post-quantum standards, CRYSTALS-Kyber and CRYSTALS-Dilithium. The authors critically re-evaluate the practical security of the Number Theoretic Transform (NTT), a core mathematical operation used to accelerate polynomial multiplication in both algorithms. While the NTT has long been recognized as a primary target for Side-Channel Attacks (SCA), traditional approaches like Correlation Power Analysis (CPA) often suffer from high operational costs, requiring the recovery of all secret key coefficients (e.g., all 256 coefficients in the NTT domain) and demanding a large number of power traces. This paper's central thesis is that such a complete and data-intensive side-channel recovery is unnecessary. The authors instead demonstrate that a partial key recovery via SCA, when combined with a powerful lattice-based algebraic solver, can achieve a complete key-recovery attack with significantly greater speed and efficiency.

The core contribution of the work is the novel "SIS-assisted method," which bridges the gap between physical side-channel leakage and the underlying mathematical structure of the algorithms. The attack's name, "When NTT Meets SIS," refers to this hybrid strategy: the attacker first uses an efficient, parameter-adjustable CPA to recover only a small *subset* of the private key's coefficients in the NTT domain. Instead of continuing the side-channel attack to

find the remaining coefficients, the authors formulate the problem of recovering the *full* key from this *partial* information as an instance of the Short Integer Solution (SIS) problem. By skillfully embedding the NTT transformation matrix into an SIS search problem, they leverage standard lattice-reduction algorithms (like BKZ) to algebraically and rapidly solve for the rest of the key.

This two-stage methodology proves to be exceptionally practical. The paper details how the SIS-assisted method dramatically reduces the requirements for a successful attack. For instance, in the case of Dilithium-2 and Kyber-512, which both have 256-coefficient private key polynomials, the authors demonstrate that they only need to recover a partial set of 32 and 64 coefficients, respectively, through the side-channel attack. This partial recovery is achieved by a "parameter-adjustable CPA" scheme, which is optimized to expedite the recovery of even a single coefficient in the NTT domain. Once this minimal threshold of information is obtained from the physical device, the attack moves entirely to a software-based lattice solver, which efficiently deduces the complete private key.

To validate their findings, the authors conducted practical attacks on both unprotected and masked implementations of Kyber and Dilithium running on a standard ARM Cortex-M4 microcontroller. The results are striking: for Dilithium-2, they achieved full private key recovery in 0.5 minutes using just 2,000 power traces. For Kyber-512, the attack was even faster, requiring only 0.4 minutes and 500 power traces. These metrics represent a significant reduction in both data complexity and computational time compared to previously known SCAs on the NTT, effectively lowering the barrier for practical physical attacks against these standardized algorithms. In conclusion, this paper presents a formidable new class of hybrid attack that cleverly combines the strengths of physical side-channel analysis and algebraic cryptanalysis. By demonstrating that only a fraction of the secret key needs to leak physically, the authors challenge the security assumptions of implementations that may have been considered "partially secure" or "too difficult" to attack with traditional SCA. The work underscores the critical need for robust, full-scale countermeasures and highlights that even minor or partial information leakage from the NTT can be deterministically amplified into a full key-recovery attack when combined with the power of lattice-based solvers.

2.3 Breaking FALCON Post-Quantum Signature Scheme through Side-Channel Attacks [3]

This paper by Aysu et al. provides the first practical side-channel analysis of the FALCON post-quantum digital signature scheme, which was a finalist in the NIST standardization process. Unlike its fellow lattice-based finalists, Kyber and Dilithium, which rely on arithmetic in finite fields using the Number Theoretic Transform (NTT), FALCON is distinguished by its unique construction based on NTRU lattices. This design necessitates polynomial arithmetic over rational numbers, which its designers implemented using floating-point operations within a Fast Fourier Transform (FFT). This unique design choice was believed to offer efficiency and compactness, but its resilience against physical attacks was largely unstudied. This paper demonstrates that this specific design choice is, in fact, a significant implementation vulnerability, and exploits it to mount a practical attack that recovers the secret key.

The authors identify the core vulnerability within the FALCON signing procedure. Specifically, the attack targets the multiplication of a secret key polynomial (e.g., polynomial 'f') with a known, public polynomial ('c') derived from the hashed message. Because this computation is performed in the FFT domain using floating-point numbers, the physical implementation (e.g., a microcontroller) leaks data-dependent information via its power consumption or, more notably in this work, its electromagnetic (EM) emissions. The authors recognized that a direct, conventional side-channel attack like Differential Power Analysis (DPA) would be insufficient. The complex nature of floating-point arithmetic with its separate sign, exponent, and mantissa components would lead to an unmanageable number of "false positives" in the analysis, making it impossible to distinguish correct key guesses from noise.

To overcome this, the paper's main contribution is a novel "extend-and-prune" attack methodology tailored specifically for FALCON's floating-point arithmetic. This is a known-plaintext attack where the adversary captures EM measurements while the device signs a number of known messages. The attack works in a multi-stage process. First, the "extend" phase involves making initial hypotheses about the secret key and identifying all potential matches in the side-channel data, even those that are ambiguous or conflicting. Then, the "prune" phase introduces a sophisticated filtering strategy. By cross-correlating information from different measurements and leveraging the algebraic properties of the algorithm.

The practical results of this attack are highly effective and demonstrate a significant threat to unprotected implementations of FALCON. The authors implemented their attack on the official reference software provided in the NIST submission package, running on a standard embedded device. They show that a complete recovery of the secret signing key is possible using approximately 1,000 electromagnetic (EM) traces (i.e., measurements from 1,000 signing operations). Once the secret key is recovered, the attacker can forge valid signatures on any arbitrary message, completely breaking the security of the scheme.

In conclusion, this work was a landmark contribution to the security analysis of post-quantum cryptography. It was the first paper to demonstrate a practical side-channel attack on the FALCON signature scheme, proving that its unique reliance on floating-point FFT—a departure from the more common NTT—was not a barrier to physical attacks but rather a distinct and exploitable vulnerability. The paper's novel "extend-and-prune" strategy provided a new tool for analyzing complex arithmetic, and its findings underscored the critical need for algorithm-specific countermeasures in all PQC implementations, irrespective of their underlying mathematical structure.

2.4 Single-Trace Side-Channel Attacks on CRYSTALS-Dilithium [4]

This paper by Liu et al. presents a significant and alarming advancement in the practical security analysis of CRYSTALS-Dilithium, the digital signature algorithm selected by NIST as a primary post-quantum standard. While many side-channel analyses focus on statistical attacks like Correlation Power Analysis (CPA), which require the collection of thousands of power traces, this work demonstrates the feasibility of a far more potent threat: a single-trace attack. Such an attack, which recovers the entire secret key from just *one* signature operation, represents a critical real-world threat model, as an attacker may only have a single opportunity to capture a device's emissions (e.g., during a single "sign-on" event). The authors successfully mount such an attack by targeting the core polynomial multiplication routines within Dilithium's signing process.

The fundamental vulnerability exploited in this work stems from the direct, data-dependent power consumption of the microcontroller executing the Dilithium algorithm. The signing procedure of Dilithium involves the computation of polynomial operations, most notably the multiplication of the secret key polynomials (s1, s2) with a public polynomial c derived from the

message hash. In a typical, unprotected implementation on an embedded device, these polynomial multiplications are executed in a sequential loop. The authors show that the power consumed by the device during each iteration of this loop is correlated with the specific values of the secret key coefficients being processed. This is a classic Simple Power Analysis (SPA) vulnerability, where the attacker does not need statistical averaging across many traces but can instead directly read the secret from the visible, data-dependent patterns within a single trace.

The authors' methodology provides a practical template for this attack. They first conduct a profiling phase on a test device to build a precise "template" of the power consumption for different operations and data values. Once this template is built, the attack phase is devastatingly simple. The attacker captures a single power trace of a signing operation from a target device (which uses the same hardware). By meticulously analyzing this trace, they can identify the exact "window" of time where the critical polynomial multiplication occurs. Then, by stepping through this window and comparing the observed power consumption patterns against their pre-built templates, they can sequentially recover each coefficient of the secret key polynomial s1. The attack is then repeated on the multiplication involving s2, leading to a full recovery of the signing key.

To validate their attack, the paper details a successful implementation against a standard, unprotected version of CRYSTALS-Dilithium running on a widely-used ARM Cortex-M4 microcontroller, a common platform for embedded systems and IoT devices. Their experimental results confirm that a complete recovery of the secret key is not just a theoretical possibility but a practical reality. This finding is particularly damaging because it bypasses the algorithm's inherent rejection sampling mechanism a probabilistic technique used in Dilithium to prevent other types of (non-physical) attacks. The side-channel leakage occurs during the computation *before* a potential rejection, meaning the attacker can extract the key regardless of whether the final signature is accepted or rejected in that attempt.

In conclusion, this paper serves as a critical warning to the cryptographic community and hardware implementers. It shatters any assumption that the algorithmic complexity of lattice-based schemes like Dilithium provides inherent protection against single-trace side-channel attacks. The work demonstrates that without explicit and robust countermeasuressuch as masking (which splits secrets into random shares) or shuffling (which randomizes the order of operations) implementations of Dilithium remain critically vulnerable to a very powerful and practical attack class.

2.5 Profiling Side-Channel Attacks on Dilithium: A Small Bit-Fiddling Leak Breaks It All [5]

This paper by H. Steffen et al. delivers a powerful and precise demonstration of the fragility of modern cryptographic implementations, focusing on the NIST-selected standard CRYSTALS-Dilithium. The work moves into the domain of high-precision profiling attacks a sophisticated class of side-channel attacks (like template attacks) where the adversary first characterizes an identical "profiling" device to build a highly accurate model of its physical leakages. The authors' central thesis is that a complete break of Dilithium is possible by exploiting minuscule, previously overlooked leakages emanating from low-level, internal "bit-fiddling" operations, specifically within the Number Theoretic Transform (NTT), which is the core engine for polynomial multiplication.

The attack's novelty lies in what it targets. Instead of seeking a large, obvious leakage pattern corresponding to a full byte or word of the secret key, the authors focus on the tiny, almost-imperceptible physical emissions from fundamental, bit-level operations. These "bit-fiddling" operations such as bit shifts, comparisons, or conditional branches are integral to the optimized implementation of the NTT's butterfly units and modular reduction steps. While these operations might seem inconsequential, they are still data-dependent, and this paper proves they leak just enough information for a profiled attack to succeed.

The methodology is divided into two distinct phases. First, in the profiling phase, the attackers use a device under their full control, identical to the target. They execute the Dilithium signing operation thousands of times with known keys and inputs, meticulously capturing power or electromagnetic (EM) traces. Using machine learning and statistical analysis, they build a precise template that models the exact leakage of these internal "bit fiddling" operations. This model learns to perfectly distinguish, for example, the tiny difference in power consumption when a single bit being processed is a '0' versus a '1'.

Second, in the attack phase, the adversary only needs to capture a very small number of traces, perhaps even just one from the victim device. They apply their hyper-specific template to this captured trace. By identifying the leakage from the profiled bit-fiddling operations as they occur during the multiplication of the secret key polynomial, the attacker can recover partial information about the key.

In conclusion, this work serves as a stark reminder of the challenges in building secure hardware. It demonstrates that for a sophisticated adversary, there is no such thing as a "safe" or "minor" information leak. By leveraging the power of profiling attacks, the authors show that even the most fundamental, single-bit operations within an algorithm's core can be turned into a catastrophic vulnerability. This finding places immense pressure on implementers, reinforcing that masking a countermeasure that splits secrets into multiple random shares to break this data-dependency is not optional but an absolute necessity for any secure implementation of CRYSTALS-Dilithium.

2.6 Exploiting Small-Norm Polynomial Multiplication with Physical Attacks: Application to CRYSTALS-Dilithium [6]

This paper by Bronchain et al. introduces a novel and highly effective physical attack vector against CRYSTALS-Dilithium, the NIST-selected standard for post-quantum digital signatures. The attack's ingenuity lies in its exploitation of a fundamental mathematical design choice that underpins Dilithium's security and efficiency: the use of "small-norm" polynomials for the secret key. In Dilithium, the secret key polynomials are not composed of uniformly random coefficients. Instead, their coefficients are drawn from a very narrow, centered, and non-uniform distribution (e.g., integers between $-\eta$ and $+\eta$, where η is a small integer like 2 or 4). This "small-norm" property is essential for the scheme's correctness and its theoretical security. This paper, however, demonstrates that this very property creates a new and potent vulnerability to physical attacks.

The authors' central thesis is that this non-uniform distribution of secret data provides a powerful "distinguisher" for a side-channel adversary. In a typical side-channel attack, the adversary must distinguish between all possible values of a secret (e.g., 0 through 255 for a byte). Here, the adversary knows the secret coefficients are not uniformly random; they are heavily biased towards zero and a few small surrounding values. The attack, therefore, targets the core signing operation where a public polynomial (derived from the message) is multiplied by these "small-norm" secret polynomials. The physical leakage (e.g., power consumption or electromagnetic emissions) of the device when it computes, for example, public * 0 or public * 1 is characteristically different and more distinct than it would be if the secret coefficients were large or uniformly random.

The paper presents a methodology that leverages this statistical bias to great effect. The authors likely employed a profiling attack (such as a template attack or a machine-learning-based approach). In a profiling phase, they first use a copy of the target device to build a highly accurate model of its physical leakages. This model is not just trained to recognize the value of the secret, but rather to recognize the distinctive statistical signature of a "small-norm" multiplication. This makes the attack exceptionally potent, as the adversary is not searching for a "needle in a haystack" (a random value) but is instead looking for a very specific, known, and biased signal.

The practical results presented in the paper demonstrate the efficacy of this approach. By targeting the implementation of this small-norm polynomial multiplication, the authors show that they can significantly reduce the complexity required to mount a full key recovery. The attack is more efficient and requires fewer traces than a generic attack that does not take this "small-norm" property into account. This work effectively turns a core, algorithm-level design feature into a concrete, implementation-level vulnerability.

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In conclusion, this paper provides a critical insight into the subtle interplay between the mathematical design of post-quantum algorithms and their physical security. It proves that security assumptions must be re-evaluated in the context of physical implementation. The authors demonstrate that the "small-norm" property, which is integral to Dilithium's theoretical security framework, paradoxically provides a powerful "tell" for side-channel adversaries. This finding strongly reinforces the non-negotiable requirement for robust countermeasures, particularly masking, in all PQC implementations.

2.7 Machine Learning Based Blind Side-Channel Attacks on PQC-Based KEMS A Case Study of Kyber KEM [7]

This invited paper by P. Ravi et al. marks a significant escalation in the physical threat landscape for post-quantum cryptography, focusing on the NIST-selected standard, CRYSTALS-Kyber (KEM). The work moves beyond traditional, statistically-based side-channel attacks (SCAs) and demonstrates the formidable power of Machine Learning (ML) and Deep Learning (DL) models to execute "blind" attacks. A blind attack is a particularly dangerous and realistic threat model where the attacker has no prior "profiling" device to characterize. Instead of building a perfect leakage template beforehand, the adversary uses the power of ML to learn the leakage model directly from the victim device, often with minimal prior knowledge. This paper's case study on Kyber proves that this advanced approach is not just theoretical but highly practical.

The central and most alarming contribution of this work is its demonstration that modern ML/DL models can magnify and exploit extremely low-order leakages. In the context of cryptographic implementations, "low-order" leakage refers to very small, subtle, and noisy information leaks that are often considered "safe" or at least "too difficult" to exploit with conventional statistical attacks like Correlation Power Analysis (CPA). These low-order leakages might be the residual, imperfectly-masked signals from an implementation that has some basic, first-order countermeasures. This paper shatters that assumption of safety, showing that advanced neural networks are exceptionally adept at finding and amplifying these faint, complex, and non-linear patterns from a noisy signal.

The paper's methodology is presented as a case study on the Kyber Key Encapsulation Mechanism. The authors likely targeted a core component of the Kyber decapsulation process, such as the Number Theoretic Transform (NTT) or the subsequent modular arithmetic and comparison steps, where the secret key is directly used. They then applied a sophisticated ML/DL framework (such as a Convolutional Neural Network (CNN), which is well-suited for time-series signal analysis) to power or electromagnetic (EM) traces collected from a device running Kyber. The "blind" nature of the attack means the ML model was able to learn the device's specific leakage characteristics and correlate them to secret-dependent operations without a pre-built profile.

The practical validation, likely performed on a common ARM Cortex-M4 microcontroller, would have confirmed the success of this ML-based blind attack. The results demonstrate that even if an implementation is protected by a low-order countermeasure (e.g., an imperfectly implemented first-order masking scheme), the ML model can still learn to break the key, effectively rendering that countermeasure useless. The attack's success underscores that the pattern-recognition capabilities of AI have outpaced the assumptions of traditional side-channel security.

In conclusion, this paper serves as a critical and timely warning. It bridges the gap between the rapidly advancing fields of artificial intelligence and hardware security, proving that ML is not just a tool for analysis but a powerful weapon for "blind" attacks. It fundamentally raises the bar for secure PQC implementation, making a compelling case that "good enough" or "low-level" countermeasures are no longer sufficient. This work strongly implies that only provably secure, high-order, and robust countermeasures designed to withstand the analytical power of modern ML are viable for protecting the next generation of standardized cryptography.

2.8 Magnifying Side-Channel Leakage of Lattice-Based Cryptosystems with Chosen Ciphertexts: The Case Study of Kyber [8]

This paper by Z. Xu et al. presents a novel and powerful attack methodology that combines the principles of active chosen-ciphertext attacks with machine learning-based profiling to break the physical implementations of lattice-based cryptosystems. The authors' work provides a practical case study against CRYSTALS-Kyber, the key encapsulation mechanism (KEM) selected by NIST as a primary post-quantum standard. The central thesis is that an adversary can actively *magnify* or amplify the side-channel leakage of a device by feeding it specific, maliciously-crafted ciphertexts. This amplified leakage, when analyzed by a potent machine learning model, leads to a highly effective and efficient key recovery.

This attack represents a significant departure from typical passive side-channel analyses. In a passive attack, the adversary merely observes the device as it processes valid, randomly-generated ciphertexts. In contrast, this paper's methodology is *active*. The adversary leverages the ability to send any ciphertext to the Kyber decapsulation oracle. They do not send random ciphertexts; they send "chosen ciphertexts" that are specially engineered to trigger specific, high-contrast internal states within the algorithm. For example, a carefully crafted

ciphertext might be designed to force an intermediate comparison or calculation to involve a value of exactly zero, while another ciphertext forces it to be a large, non-zero value. This ability to control the internal state creates a much "louder" and more distinct leakage signal for the targeted operation, effectively "magnifying" the side-channel emissions.

The paper's methodology then combines this amplification technique with the power of machine learning-based profiling. The attack consists of two phases. First, in the profiling phase, the attacker uses an identical "training" device. They feed this device their "magnifying" chosen ciphertexts and use a machine learning model (such as a convolutional neural network) to learn the precise, amplified leakage patterns associated with secret-dependent operations. Because the leakage is intentionally magnified, the ML model can be trained with high precision and requires less data to learn the distinguishing features.

Second, in the attack phase, the adversary targets the victim device. By feeding it the same set of chosen ciphertexts and applying the trained ML model to the captured traces, the attacker can recover the secret key. The authors' case study on Kyber demonstrates the practical success of this combined approach. The attack likely targets the core of the decapsulation process where the received ciphertext is used to re-compute a shared secret, a step that is critically dependent on the private key. This work is particularly notable because while Kyber is *theoretically* secure against chosen-ciphertext attacks (IND-CCA2), this paper demonstrates that its *physical implementation* can still be broken by an adversary who uses chosen ciphertexts to amplify physical, non-theoretical leakage.

In conclusion, this paper provides a crucial insight into the security of PQC implementations. It proves that an active attacker who can control the inputs to a device is a far greater threat than a passive one. By demonstrating that chosen ciphertexts can be used to "amplify" leakage, The adversary leverages the ability to send any ciphertext to the Kyber decapsulation oracle. They do not send random ciphertexts; they send chosen ciphertexts that are specially engineered to trigger specific, high-contrast internal states within the algorithm the authors show that standard countermeasures designed to protect against passive attacks may be insufficient. This work effectively bridges the gap between active fault injection and passive side-channel analysis, reinforcing the absolute necessity for robust, multi-layered defenses that protect against all forms of physical attacks.

2.9 More Practical Single-Trace Attacks on the Number Theoretic Transform [9]

This paper by Pessl and Primas (2019) represents a pivotal moment in the security analysis of lattice-based cryptography, specifically targeting the Number Theoretic Transform (NTT). The NTT is the computational heart of many efficient post-quantum schemes, including the NIST-selected standard Kyber. Prior to this work, single-trace side-channel attacks on the NTT were known but considered largely impractical; they either targeted "variable-time" implementations (which are already known to be insecure) or required an exorbitant amount of profiling effort (e.g., millions of templates) to succeed against constant-time code. This paper challenges that status quo, demonstrating that with improved methodologies, single-trace attacks on highly optimized, constant-time implementations are not only possible but "more practical" than previously assumed.

The authors' primary contribution is a refined and highly efficient attack framework that combines template-based profiling with Soft-Analytical Side-Channel Attacks (SASCA) using Belief Propagation (BP). The attack targets the modular multiplications within the "butterfly" operations the atomic steps of the NTT. In a traditional template attack, an adversary profiles a device to learn the leakage of specific intermediate values. However, in the context of the NTT, the sheer number of possible inputs and intermediates makes building a template for every possibility computationally prohibitive. The authors overcome this by leveraging the structured nature of the NTT. They model the entire NTT computation as a factor graph—a probabilistic map where nodes represent intermediate variables and edges represent the mathematical operations connecting them.

Instead of trying to recover the key from a single point of leakage, the authors use Belief Propagation to pass information (probabilities) across this graph. If the template attack gives a "weak" guess for one intermediate value, the BP algorithm can refine that guess by propagating information from neighbors in the graph (e.g., "if the output of this butterfly is likely X and one input is likely Y, then the other input must be Z"). This holistic approach allows the attacker to exploit leakage from every step of the NTT simultaneously, accumulating enough information from a single execution trace to recover the inputs.

A key strategic shift in this paper is the authors' decision to target the encryption procedure rather than the decryption procedure. While decryption involves the long-term private key, encryption involves the generation of ephemeral secrets (random "noise" polynomials). Recovering these ephemeral secrets allows the attacker to mathematically derive the shared session key or, in some contexts, the long-term key. The authors argue that encryption is often less protected than decryption, and crucially, the "noise" polynomials processed during encryption have values drawn from a very narrow range (small integers). This "small value" property drastically reduces the search space for the template attack, allowing the authors to succeed with as few as 213 templates, massive reduction from millions required in prior works.

To validate their findings, the authors performed practical attacks on a Cortex-M4 microcontroller running a constant-time implementation of Kyber. The results were decisive, they demonstrated that a single power trace of the NTT operation was sufficient to recover the ephemeral keys with high probability. This effectively bypasses the standard "constant-time" defense, which protects against timing attacks but not against power analysis.

In conclusion, this paper significantly lowered the bar for physical attacks on post-quantum cryptography. It dispelled the notion that constant-time implementations are inherently safe from single-trace analysis without dedicated masking. By demonstrating that advanced statistical techniques (SASCA) can be combined with smart target selection (encryption/small-noise), the authors proved that the NTT the very engine of efficiency for lattice schemes is also a highly sensitive breakage point that demands rigorous, masked protection in all practical deployments.

2.10 Carry Your Fault: A Fault Propagation Attack on Side-Channel Protected LWE-based KEM [10]

This paper by Kundu et al. (2023) introduces a sophisticated and paradoxical new attack vector against post-quantum cryptography, specifically targeting the NIST-selected standard CRYSTALS-Kyber. The work is significant because it identifies a critical vulnerability not in the cryptographic algorithm itself, but within a standard countermeasure designed to protect it. The authors demonstrate that the Arithmetic-to-Boolean (A2B) conversion gadget a component widely introduced into implementations to enable side-channel masking can be exploited via fault injection to recover the secret key.

The core of the attack lies in the data-dependent nature of the carry chain within the A2B conversion process. In a masked implementation of a lattice-based KEM like Kyber, sensitive values are split into random shares to prevent passive side-channel analysis (SCA). Mathematical operations often require switching these shares between arithmetic masking (modulo q) and Boolean masking (XOR-based). The A2B gadget performs this conversion, typically using an adder circuit. The authors observe that the propagation of the "carry" bit in this adder is highly sensitive to the input data. By injecting a precise fault specifically targeting the arithmetic shares just before this conversion the attacker can disturb the carry propagation. The methodology employs a "Fault Propagation" strategy combined with an inequality-solving framework. The attack proceeds as follows:Fault Injection: The adversary injects a controlled fault (e.g., a "stuck-at" fault) into one share of the secret data during the decapsulation process.

Carry Propagation Analysis, This fault propagates through the A2B adder's carry chain. Crucially, whether the fault propagates to higher bits or is "absorbed" depends on the specific values of the secret shares and the noise term involved in the LWE computation. Oracle Feedback, The attacker observes whether the decapsulation succeeds or fails (a standard oracle response in KEMs). A failure indicates that the fault propagated significantly enough to corrupt the message, while success indicates the fault was absorbed or had minimal impact.

Inequality Generation Each success/failure response provides a mathematical inequality involving the secret key and the noise. Because the noise is small and bounded (a feature of LWE), these inequalities leak information about the secret key. Belief Propagation (BP), The authors utilize a Belief Propagation algorithm to efficiently solve the system of generated inequalities. By accumulating enough faulted traces, the BP solver converges on the correct secret key.

To validate the threat, the authors performed practical experiments on a masked implementation of Kyber running on a standard ARM Cortex-M4 microcontroller (STM32F4). They successfully demonstrated key recovery using Electromagnetic (EM) fault injection. The results are particularly alarming because the attack works against implementations that are theoretically secure against passive side-channel attacks (due to masking) and generic fault attacks. The authors show that the very mechanism intended to hide the key (the A2B conversion) becomes the instrument of its exposure when subjected to active faulting.

In conclusion, "Carry Your Fault" exposes a dangerous interplay between different security countermeasures. It proves that "hardening" an implementation against one threat (Side-Channel Analysis) by adding complex conversion gadgets can inadvertently lower the bar for another threat (Fault Injection). The paper argues that future PQC implementations must not just apply masking and fault countermeasures in isolation but must fundamentally redesign conversion gadgets (like A2B) to be fault-aware or "fault-absorbent," ensuring that internal carry chains do not become side-channels for active adversaries.

2.11 Cache-timing Attack Detection and Prevention Application to Crypto Libs and PQC [11]

As the threat of cache-timing attacks continues to evolve, particularly with the rise of microarchitectural vulnerabilities like Spectre and Meltdown, the ability to verify the security of cryptographic software has become critical. This paper by S. Carré et al. (2018/2019) addresses the disconnect between theoretical "vulnerabilities" (patterns in source code that *might* leak) and practical "exploitations" (actual successful attacks). The authors introduce a rigorous, automated methodology to characterize and detect information leakage in cryptographic libraries, bridging the gap between static code analysis and dynamic attack realization. Their work is particularly notable for applying these techniques not only to established libraries like MbedTLS but also to the emerging field of Post-Quantum Cryptography (PQC), demonstrating that the "next generation" of cryptographic algorithms is not immune to classical implementation flaws.

The core contribution of the paper is the proposal of a detection tool and methodology, referred to in the study as "Stanalyzr" (and associated with the commercial tool Catalyzr). The authors employ a white-box static analysis approach that operates on the Abstract Syntax Tree (AST) of the source code. The methodology is built on the concept of "taint analysis." It identifies "sensitive variables"—those that depend on secret keys and tracks their propagation through the code. The tool specifically flags two types of critical vulnerabilities, Control-Flow Leakage, Instances where a sensitive variable determines the execution path (e.g., a secret-dependent if condition or loop bound). This leads to timing differences based on the instructions executed. Data-Flow Leakage, Instances where a sensitive variable is used as an index to access memory (e.g., Table[secret]).

This leads to cache-timing leaks (like Flush+Reload or Prime+Probe) because the memory access pattern reveals the secret. To validate their methodology, the authors applied their tool to widely used cryptographic libraries. A key case study presented in the paper is the analysis of MbedTLS (specifically version 2.14.0). The tool successfully identified previously unknown vulnerabilities in the RSA signature implementation, where secret-dependent operations were found to influence the execution flow. Furthermore, the paper extends its scope to PQC, acknowledging that while the underlying mathematics of schemes like DAGS (a code-based KEM) or lattice-based algorithms may be quantum-resistant, their software implementations often rely on large lookup tables or complex conditional logic that make them highly susceptible to cache-timing attacks.

The authors demonstrate that their automated analysis can pinpoint these non-constant-time constructs in PQC reference implementations, which are often written with a focus on performance rather than side-channel resistance. Beyond detection, the paper discusses prevention strategies. It argues that the traditional reliance on "constant-time" programming (avoiding secret-dependent branches and lookups) is difficult to achieve and verify manually.

The authors propose and evaluate automated countermeasures, such as efficient window-based techniques, to eliminate "vertical" leakage (leakage within a single execution) without incurring the prohibitive performance costs often associated with full masking or blinding.

The paper extends its scope to PQC, acknowledging that while the underlying mathematics of schemes like DAGS or lattice-based algorithms may be quantum-resistant. The tool specifically flags two types of critical vulnerabilities, Control-Flow Leakage, Instances where a sensitive variable determines the execution path.

In conclusion, this paper provides a vital framework for the security assurance of modern cryptographic software. By automating the detection of cache-timing vulnerabilities via AST-based taint analysis, the authors offer a scalable solution to a pervasive problem. Their application of this tool to both standard libraries (MbedTLS) and PQC schemes highlights a universal truth in cryptography, a mathematically secure algorithm is only as strong as its implementation. The work underscores the urgent need for automated verification tools in the PQC standardization process.

2.12 Investigating CRYSTALS-Kyber Vulnerabilities: Attack Analysis and Mitigation [12]

This paper, published in 2024, presents a rigorous empirical investigation into the physical security of CRYSTALS-Kyber, the lattice-based algorithm recently standardized by NIST as the primary Key Encapsulation Mechanism (KEM) for the post-quantum era. While Kyber offers strong theoretical guarantees against quantum adversaries, its resistance to implementation-level attacks specifically Side-Channel Attacks (SCA) remains a critical area of concern. The authors of this study address this gap by conducting a comprehensive attack analysis on reference implementations of Kyber (specifically Kyber-512), focusing on the efficacy of existing countermeasures and revealing significant residual vulnerabilities that persist even in the presence of standard defense mechanisms.

The core of the research focuses on the susceptibility of Kyber to Chosen-Ciphertext Attacks (CCA) assisted by side-channel leakage. The authors demonstrate that an attacker who can manipulate the input ciphertext even within the bounds of validity checked by the protocol an induce specific, data-dependent leakage patterns in the device's power consumption or electromagnetic emissions. The study specifically targets the decapsulation phase of Kyber, identifying it as the most vulnerable component. During decapsulation, the device performs critical operations involving the long term secret key, such as the Number Theoretic Transform (NTT) and the message decoding steps. The authors show that by feeding "selected" or maliciously crafted ciphertexts to the device, they can trigger specific execution paths or intermediate value states that leak highly correlated information about the secret key.

A pivotal contribution of this work is its critical evaluation of countermeasures. The authors investigate the effectiveness of standard mitigation techniques, such as masking (splitting sensitive data into random shares) and shuffling (randomizing the order of operations), which are commonly proposed to secure Kyber. The paper reveals a concerning "efficacy gap." It demonstrates that while these countermeasures increase the difficulty of an attack (requiring more traces or higher-order analysis), they do not render the implementation immune. The study highlights that sophisticated attack vectors, particularly those leveraging chosen ciphertexts to amplify leakage (a technique often called "ciphertext malleability" or "leakage amplification"), can bypass these defenses.

Furthermore, the paper delves into the implementation-specific flaws that facilitate these attacks. It points out that "additional functions" within the reference implementation often auxiliary routines used for memory management, data movement, or specific modular reductions can become inadvertent leakage sources. These secondary functions are frequently overlooked by developers who focus their hardening efforts solely on the primary arithmetic operations (like the NTT), leaving "side doors" open for attackers. The authors use this finding to argue that security must be holistic; hardening only the mathematically complex parts of the algorithm is futile if the surrounding "glue code" leaks the secret.

In conclusion, this research serves as a wake up call for the post-quantum cryptography community. It challenges the assumption that simply porting standard masking techniques to Kyber is enough to guarantee physical security. By demonstrating the "inefficacy" of these measures against optimized chosen ciphertext side channel vectors, the authors underscore the need for a new generation of countermeasures. They suggest that future implementations must integrate more robust, potentially hardware assisted defenses or advanced, AI-driven anomaly detection systems to identify and mitigate these subtle, amplified leakages before they can be exploited for full key recovery. The paper ultimately reinforces that as Kyber moves from a research candidate to a global standard, its implementation security must evolve to withstand not just theoretical quantum attacks, but the gritty, practical reality of physical side-channels.

2.13 Defeating low-cost countermeasures against side-channel attacks in lattice-based encryption: A case study on Crystals-Kyber [13]

This paper by P. Ravi et al. (2023) presents a critical security evaluation of "low-cost" defense mechanisms proposed for post-quantum cryptography. As the computational overhead of robust side-channel countermeasures (like high-order masking) can be prohibitive for resource-constrained devices, researchers have explored alternative, lightweight "detection-based" countermeasures. These mechanisms do not try to prevent leakage mathematically but instead aim to *detect* and block the specific input patterns used by adversaries, such as the malicious, low-entropy ciphertexts typical of chosen-ciphertext attacks. This paper systematically dismantles this defensive philosophy, using the NIST-standardized CRYSTALS-Kyber KEM as a case study.

The authors presents a critical security evaluation of "low-cost" defense mechanisms proposed for post-quantum cryptography. As the computational overhead of robust side-channel countermeasures (like high-order masking) can be prohibitive for resource-constrained devices, researchers have explored alternative, lightweight "detection-based" countermeasures. These mechanisms do not try to prevent leakage mathematically but instead aim to *detect* and block the specific input patterns used by adversaries, such as the malicious, low-entropy ciphertexts typical of chosen-ciphertext attacks. This paper systematically dismantles this defensive philosophy, using the NIST-standardized CRYSTALS-Kyber KEM as a case study to demonstrate that such detection-based protections are fundamentally insufficient against adaptive adversaries.

The authors focus their analysis on two specific classes of low-cost countermeasures: Ciphertext Sanity Checks and Decapsulation Failure Checks. The "Sanity Check" defense operates on the premise that most side-channel attacks require the adversary to send "crafted" ciphertexts that are mathematically distinct from normal, random traffic—often containing many zeros or repeating patterns (low entropy) to simplify the attack. The countermeasure simply inspects incoming ciphertexts and rejects those that appear"too simple or malicious. The "Decapsulation Failure Check" operates on the assumption that attacks often induce decryption failures to gain information; thus, the device should halt or reset if too many failures occur.

The paper introduces novel attack strategies to bypass both defenses completely. To defeat the Ciphertext Sanity Check, the authors propose a technique to "mask" their malicious inputs. They demonstrate that an attacker can take a standard, low-entropy attack ciphertext (which would normally be detected) and mathematically "blind" it using the target's public key. This transformation effectively "whitens" the data, increasing its entropy so that it appears indistinguishable from a valid, random ciphertext to the detector. Crucially, the authors prove that despite this transformation, the ciphertext retains its malicious properties during the internal decryption process, allowing the side-channel leakage to occur just as effectively as before. This renders the sanity check useless, as the attack traffic effectively "camouflages" itself as legitimate traffic. The results were conclusive the authors successfully recovered the full secret key in both scenarios. The countermeasure simply inspects incoming ciphertexts and rejects those that appear.

To defeat the Decapsulation Failure Check, the authors introduce a groundbreaking valid-ciphertext side-channel attack. Traditionally, chosen-ciphertext SCAs rely on inducing failures or specific error states. In this work, the authors develop a method to recover the secret key using *only* valid ciphertexts that successfully pass the integrity checks and complete the decapsulation process without error. Because no failure is triggered, the "Decapsulation Failure Check" is never activated. To make this possible, the authors developed an improved algebraic solver (outperforming standard Belief Propagation techniques) capable of extracting secret information from the subtle leakage of valid executions. This solver treats the leakage as a system of linear inequalities involving the secret key and the noise terms, solving them efficiently to recover the key.

The attacks were practically validated on an ARM Cortex-M4 microcontroller, targeting both unprotected (pqm4) and masked (mkm4) implementations of Kyber. The results were conclusive: the authors successfully recovered the full secret key in both scenarios, demonstrating that the low-cost countermeasures provided no meaningful security enhancement against their advanced attack vectors.

The countermeasure simply inspects incoming ciphertexts and rejects those that appear too simple or malicious. The Decapsulation Failure Check operates on the assumption that attacks often induce decryption failures to gain information; thus, the device should halt or reset if too many failures occur. These mechanisms do not try to prevent leakage mathematically but instead aim to detect and block the specific input patterns used by adversaries, such as the malicious, low-entropy ciphertexts typical of chosen-ciphertext attacks.

In conclusion, this paper serves as a definitive refutation of "security through detection" for lattice-based cryptography. It argues that attempting to "filter" malicious inputs is a losing battle, as adversaries can simply adapt their inputs to bypass the filters while maintaining the attack's lethality. The authors conclude that there are no shortcuts to security; robust, albeit expensive, preventative countermeasures such as high-order masking and shuffling remain the only viable solution for protecting Kyber and similar algorithms against physical attacks in the real world.

2.14 The Learning with Errors Problem (Invited Survey) [14]

This invited survey by Oded Regev (2010/2021) provides the definitive overview of the Learning with Errors (LWE) problem, a computational problem that has become the cornerstone of modern post-quantum cryptography (PQC). As the vulnerability of classical algorithms (like RSA and Elliptic Curve Cryptography) to quantum computing became apparent, the search for "quantum-hard" alternatives intensified. This survey establishes LWE as the primary candidate for this role, detailing its robust mathematical foundations, its unique hardness properties derived from high-dimensional lattices, and its incredible versatility in constructing advanced cryptographic primitives. For researchers studying PQC standards like FALCON, CRYSTALS-Kyber, and Dilithium, this document serves as the foundational text that explains these schemes are believed to be secure.

The survey begins by formally defining the LWE problem. In its simplest "search" form, LWE asks an adversary to solve a system of linear equations over a finite field, which would normally be trivial using Gaussian elimination. However, LWE introduces a crucial twist: each equation is perturbed by a small, random "error" term (typically drawn from a discrete Gaussian distribution). The presence of this noise makes the system computationally intractable to solve using standard linear algebra. The survey distinguishes between the Search-LWE (finding the secret solution) and Decision-LWE (distinguishing the noisy equations from purely random noise) variants, proving that they are computationally equivalent for appropriate parameters. This equivalence is vital for cryptography, as it allows security proofs to rely on the assumption that the LWE distribution is pseudorandom.

The central contribution of the survey and the reason for LWE's dominance in PQC is the detailed explanation of the worst-case to average-case reduction. Regev explains that solving the LWE problem on average (i.e., breaking a random instance of an encryption scheme) is at least as hard as solving the hardest instances of famous lattice problems, such as the Shortest Vector Problem (SVP) or the Shortest Independent Vectors Problem (SIVP), using a quantum computer. This is a powerful security guarantee. Unlike factoring or discrete log, where the "average" case might be easier than the "worst" case, LWE enjoys a guarantee that breaking it implies a breakthrough in solving worst-case lattice problems that have stumped mathematicians for decades.

Furthermore, the survey provides essential context for understanding the FALCON signature scheme. While FALCON is based on NTRU lattices rather than plain LWE, the survey discusses the closely related Short Integer Solution (SIS) problem and the concept of lattice trapdoors (specifically the Gentry-Peikert-Vaikuntanathan or GPV framework). Regev explains how LWE and SIS are dual to each other: while LWE involves "hiding" a secret with noise, SIS involves finding short, non-trivial solutions to linear equations. Schemes like FALCON rely on the hardness of SIS and the ability to sample short vectors from a lattice using trapdoor concepts that are rigorously formalized within the survey's broader discussion of lattice hardness. The survey elucidates the geometric properties of these lattices, explaining how the "shortness" of the error or solution vectors is the critical factor that separates a valid cryptographic state from a random one.

The document also catalogs the immense versatility of LWE. Regev outlines how this single assumption enables not just basic public-key encryption, but also advanced primitives that were previously impossible or inefficient, such as Identity Based Encryption (IBE) and Fully Homomorphic Encryption (FHE). This versatility is what makes LWE (and its structured variants) superior to other PQC families like code-based or multivariate cryptography.

In conclusion, Oded Regev's survey is the canonical reference for understanding the "quantum hardness" of the current PQC landscape. It connects the abstract geometry of high-dimensional lattices to the concrete security of protocols like FALCON and Kyber. By establishing that the security of these real-world schemes is inextricably tied to the hardness of worst-case lattice problems, the survey provides the confidence required to standardize and deploy these algorithms to protect global digital infrastructure against the looming threat of quantum computing.

2.15 On lattices, learning with errors, random linear codes, and cryptography [15]

This paper by Oded Regev (2024) is a seminal and updated survey that meticulously establishes the Learning with Errors (LWE) problem as the primary mathematical foundation for lattice-based post-quantum cryptography (PQC). The work synthesizes Regev's foundational breakthroughs, presenting a cohesive framework that connects high dimensional lattice geometry to a seemingly simple algebraic problem, ultimately proving that breaking the resulting

cryptographic schemes would require a revolutionary breakthrough in solving some of the hardest problems in classical computer science, even with a quantum computer. This equivalence is significant as it anchors the hardness of LWE to algebraic coding theory, a well-studied field of mathematics. The paper's enduring value lies in proving that LWE is uniquely positioned to offer the efficiency and security required for the next generation of digital infrastructure.

The core of the paper is dedicated to formally defining LWE. The problem challenges an adversary to solve a system of linear equations over a finite modulus, where each equation is obfuscated by adding a small, secret, random error term drawn from a specific distribution (typically a discrete Gaussian). The introduction of this "noise" transforms a trivial algebra problem into a seemingly intractable computational one. The paper rigorously details the formal definition of LWE, distinguishing between its Search and Decision variants, and outlining the precise mathematical parameters the modulus, the number of equations, and the error distribution required for achieving quantum level hardness.

A pivotal and unique aspect of this paper is the explicit connection it draws between LWE and random linear codes. The survey demonstrates that LWE is computationally equivalent to finding an arbitrary vector close to a random linear code. This equivalence is significant as it anchors the hardness of LWE to algebraic coding theory, a well-studied field of mathematics. The cryptographic security of the scheme can thus be viewed through two equivalent lenses: one based on the geometric properties of lattices (the standard view) and one based on the properties of codes. This dual perspective is a testament to LWE's inherent robustness and versatility, reinforcing the confidence in its foundational security.

The most critical result detailed in the survey is the famous worst-case to average-case reduction. Regev proves that if an adversary can find an algorithm to solve the LWE problem on average (i.e., successfully decrypt a random ciphertext), that same algorithm can be used to solve the Shortest Vector Problem (SVP) and the Shortest Independent Vectors Problem (SIVP) on any high dimensional lattice, including the worst-case instances that are believed to be classically and quantumly hard. This guarantee is the bedrock of PQC security. It means that the security of LWE-based schemes is not contingent on the randomness of the secret key or the choice of the public key, but on the intrinsic, fundamental hardness of high dimensional lattices themselves.

This framework provides the essential mathematical context for understanding the security of PQC standards. The paper's principles directly underpin the security of CRYSTALS Kyber and CRYSTALS-Dilithium, which are based on the structured LWE (Module LWE) and SIS (Short

Integer Solution) problems, the latter being the dual of LWE. The problem challenges an adversary to solve a system of linear equations over a finite modulus, where each equation is obfuscated by adding a small, secret, random error term drawn from a specific distribution.

The paper's overarching discussion of lattice hardness and the SIS problem which is central to FALCON's signing procedure provides the broader cryptographic context. The cryptographic security of the scheme can thus be viewed through two equivalent lenses one based on the geometric properties of lattices (the standard view) and one based on the properties of codes. The ability of the Regev framework to connect worst-case lattice problems to the average case security of these deployed schemes is what certifies them as viable quantum resistant alternatives. In essence, the survey proves that these schemes derive their security from the deepest mathematical structures know today, making LWE the indispensable foundation for quantum safe communication.

2.16 Pre-Quantum to Post-Quantum Cryptography Transition: Integration of AI, Quantum Computing, and Semiconductor Technology [16]

This paper by S. Das and A. Das (2024) presents a crucial, interdisciplinary analysis of the global transition from pre-quantum to post-quantum cryptography (PQC), arguing that securing this shift requires understanding the complex interplay between Artificial Intelligence (AI), Quantum Computing (QC), and the Semiconductor Technology that underlies all modern digital systems. The authors move beyond the standard PQC narrative (the simple replacement of RSA/ECC) to assess the integrated threat vectors that will inevitably arise as these technologies reach maturity, concluding that a holistic, interdisciplinary approach is essential for a successful and resilient transition.

The study first delves into the dual role of AI in the cryptographic arms race. On one hand, AI is the leading threat to cryptographic implementations. Deep Learning (DL) and Machine Learning (ML) models dramatically enhance Side-Channel Attacks (SCA), requiring significantly fewer traces for key extraction and effectively bypassing traditional, low-order countermeasures like basic masking and shuffling. AI can optimize fault injection attacks in real time and demonstrates transferability, allowing a model trained on one device to attack similar implementations on different hardware, dramatically increasing the scalability of physical attacks on PQC schemes like Kyber and Dilithium.On the other hand, AI is an essential component of the defense strategy. The authors explore AI-driven mitigation techniques, including Real-time

Anomaly Detection: AI models can continuously monitor cryptographic operations for subtle, unusual power or EM patterns, enabling the system to halt an attack in progress. Automated Countermeasure Optimization: AI and Reinforcement Learning can be used by designers to optimize complex countermeasures (like high-order masking) in hardware, balancing security assurance against performance overheads. Cryptographic Agility: AI can facilitate dynamic algorithm switching in a network, deciding when and how to transition between classical, hybrid, and PQC schemes based on real-time threat intelligence.

The paper next analyzes the fundamental shift demanded by the PQC transition at the physical layer. PQC algorithms, particularly lattice-based KEMs like Kyber, are computationally and resource-intensive compared to their classical predecessors. This necessitates a move from general-purpose CPUs to specialized hardware accelerators (ASICs, FPGAs) to achieve acceptable latency and throughput in high-volume applications like network infrastructure. The authors emphasize the intertwined relationship with semiconductor technology:

PQC in Silicon, The transition requires cryptographers to work closely with chip designers to incorporate PQC modules into the semiconductor fabrication process, ensuring these complex algorithms are resistant to both classical and quantum-accelerated physical attacks. AI for Design, AI itself is becoming critical to the semiconductor industry through Electronic Design Automation (EDA). AI tools optimize the physical layout, timing, and power consumption of PQC circuits, which is essential given the high complexity and memory demands of PQC schemes. This creates a critical circular dependency: AI is needed to efficiently build the hardware that secures against quantum threats.

The paper's core thesis centers on the emergence of integrated threat vectors that transcend simple algorithmic or physical attacks. The most dangerous threats are the synergistic combinations Quantum-Accelerated Cryptanalysis (QAC): This combines the computational power of a future quantum computer (even if small or non-perfect) with the pattern-recognition capabilities of AI. This hybrid approach could accelerate existing cryptanalysis techniques or discover unforeseen structural weaknesses in PQC algorithms faster than classical methods.

Harvest Now, Decrypt Later (HNDL) with AI: Adversaries are currently stockpiling encrypted data protected by classical cryptography. The combined power of future QC and current AI optimization models will make the eventual decryption of this harvested data faster and more scalable than previously anticipated, drastically accelerating the effective Q-Day timeline.

Side-Channel Threats: The combination of QC's potential to break symmetric keys (via Grover's algorithm) and AI's ability to exploit implementation leaks poses a continuous, integrated threat that necessitates QAI/QML-resistant PQC development. AI for Design, AI itself is becoming critical to the semiconductor industry through Electronic Design Automation.

In conclusion, the paper argues that the PQC transition is not merely a change in mathematical formulas but a radical shift in technology and threat modeling. Securing the future requires developing cryptographic agility, establishing AI security frameworks, and fostering mandatory collaboration between cryptographers, AI researchers, and semiconductor engineers to ensure that the new post-quantum landscape is secure not only against the theoretical quantum threat but also against the real-world, integrated threats that are rapidly evolving today.

2.17 RSA Key Recovery from Digit Equivalence Information [17]

This paper by C. Chuengsatiansup et al. (2021) addresses the long-standing problem of secret key recovery in classical public-key cryptography, demonstrating a powerful new cryptanalytic technique that exploits subtle, fine-grained side-channel leakages often overlooked by traditional security models. The work provides a crucial link between imperfect physical side-channel observation and sophisticated mathematical cryptanalysis, proving that even minimal, abstract information leaked by a constant-time implementation can be sufficient to compromise a large secret key. By successfully breaking RSAthe most influential public-key system—the paper reinforces the complexity of the physical threat model that must now be applied to all modern cryptography, including emerging lattice-based PQC schemes.

The Vulnerability: Digit Equivalence Leakage. The central vulnerability analyzed in the paper stems from the fixed-window exponentiation algorithm, which is commonly used in high-performance implementations of RSA, particularly the CRT-RSA variant, to accelerate decryption and signing. This algorithm breaks the large secret exponent into smaller, fixed-size digits (e.g., of window size). These digits are then processed sequentially. The paper assumes an adversary can mount a side-channel attack (like a cache-timing attack or an EM-based attack) that is precise enough to leak the equivalence between these digits, without leaking the actual value of the digits themselves. This is a highly restrictive and abstract form of leakage; the attacker only knows, for instance, that digit is the same as digit, but does not know that E[[i]] and both equal '3'.

The Methodology, Heninger-Shacham Algorithm Adaptation The primary contribution of the work is the methodology for leveraging this abstract equivalence information for a full key recovery. The authors adapt and apply the powerful Heninger-Shacham (HS) algorithm, a technique originally developed to recover RSA keys from partial, known bits (e.g., from cold boot attacks). The original HS algorithm uses a branch-and-prune approach to efficiently search the vast solution space defined by the known key bits. The paper's innovation is to model the digit equivalence leakage in a way that allows the HS algorithm to process it.

The attack proceeds in two stages:Modeling and Constraint Generation: The digit equivalence information is first modeled as a set of algebraic constraints. The total number of possible key candidates, which is computationally infeasible to brute-force. The authors mathematically model the attack as a Markov chain to analyze the probability of success and the number of operations required, showing that the equivalence information severely limits the search space.

Key Reconstruction, The constrained problem is fed into the adapted HS algorithm. By pruning the search tree based on the observed equivalences and the fundamental mathematical properties linking the secret key components (p, q, d) to the public key (N, e), the algorithm efficiently reconstructs the full RSA private key.

Results of the paper demonstrates that this cryptanalytic approach is highly feasible in the commonly used case of (a 32-digit representation of the exponent). The theoretical model predicts that approximately 65% of the keys can be broken with a search space complexity, which is well within the reach of well-resourced adversaries. The authors complement this theoretical analysis with concrete experiments on randomly generated RSA-2048 keys, validating the accuracy of their model.

In conclusion, this research provides a stark warning about the security of implementation-level details. It demonstrates that the key to key recovery is not always finding the "loudest" leakage but finding a systematic way to process any leakage, no matter how abstract. By successfully applying sophisticated algebraic techniques to a subtle physical side-channel, the paper reinforces the threat model for PQC schemes. While Kyber and Dilithium are based on different mathematics (lattices), they still rely on large integer arithmetic and fixed-window-like processing (such as the NTT). This work proves that implementing even supposedly quantum-safe algorithms with constant-time code is insufficient if small, fine-grained leakage like digit equivalence or memory access patterns can be used as inputs to powerful algebraic reconstruction algorithms.

Chapter 3
Requirement Gathering & Planning

The requirement gathering phase began with a rigorous analysis of the NIST-standardized PQC algorithms Kyber, Dilithium, and FALCON to pinpoint specific implementation level vulnerabilities. The primary technical requirement identified was the necessity to eliminate timing-based side channel leakage inherent in the default Centered Binomial Distribution (CBD) sampler. Functional requirements dictated that the replacement mechanism must maintain mathematical correctness and cryptographic equivalence to the original protocols, ensuring valid key encapsulation and signature verification without altering the final output. Crucially, the non-functional requirements emphasized strict constant-time execution, necessitating a sampling algorithm that operates with a uniform runtime profile, independent of secret input data, to effectively neutralize timing attacks.

Based on these critical requirements, the planning phase established a development roadmap centered on the integration of the Knuth-Yao (KY) sampling algorithm. The design strategy involved creating a precomputed, sparse matrix representation of the probability distribution to enable efficient, time-independent tree traversal. The project plan defined a robust technology stack, utilizing C for the core cryptographic modifications to match the reference implementations and Python for automated testing and statistical validation. A phased testing strategy was also devised, prioritizing comparative timing analysis using Welch's t-test and linear regression models. This planning ensured a structured approach to benchmarking the proposed KY solution against the baseline CBD sampler, guaranteeing that theoretical security improvements translated into practical, measurable resilience.

3.1 Software Requirements

Table II lists the essential software tools, languages, and libraries used in the implementation and analysis of the post quantum cryptographic schemes. It includes the primary programming languages for algorithm development, quantum simulation frameworks for entropy generation, and statistical libraries used to validate constant time execution. These components ensure the proposed solution is both theoretically secure and practically verifiable.

Table II: Software Requirements

Programming Language	Python
Quantum Framework	Qiskit (IBM Quantum) for circuit simulation
Analysis Libraries	NumPy, SciPy, Seaborn, Matplotlib
Performance Tools	Timeit, Perf_counter, ctypes
PQC Schemes	CRYSTALS-Kyber, CRYSTALS-Dilithium, FALCON

3.2 Functional Requirements

Table III highlights the core functionalities necessary for the modified cryptographic system to operate securely and correctly. It defines the critical operations for key management, the specific logic for side-channel resistant sampling, and the generation of quantum-safe randomness. These features ensure the system maintains cryptographic correctness while eliminating specific attack vectors.

Table III: Functional Requirements

Cryptographic Operations	Key Generation, Encapsulation (Encryption), and Decapsulation (Decryption)
Sampling Mechanism	Constant-time Knuth-Yao algorithm using sparse matrix representation
Randomness Source	Quantum-Resilient PRNG and Quantum Entropy Simulation (Hadamard/Z gates)
Verification	Bitwise identical output comparison against standard reference implementations

3.3 Non-Functional Requirements

Table IV outlines the key non-functional requirements essential for the system's security posture and operational efficiency. These attributes ensure the application resists physical attacks, performs consistently under varying inputs, and manages resources effectively. Each aspect is critical for bridging the gap between theoretical security and robust real-world deployment.

Table IV: Non-Functional Requirements

Security	Efficient timetable generation with minimal processing time
Performance	Constant-time execution profile with negligible timing variance (Low Standard Deviation)
Efficiency	Minimal computational overhead with optimized memory usage via precomputed tables
Reliability	Deterministic behavior independent of secret key data or input values

Chapter 4
Design

4.1 DFD Diagram

The Data Flow Diagrams (DFD) illustrate the secure flow of cryptographic data through the modified Post-Quantum Cryptography framework. They represent the interaction between the Sender.

4.1.1 DFD Level 0 Diagram of PQC Framework

The Level 0 Data Flow Diagram, often referred to as the Context Diagram, provides a high-level overview of the Side-Channel Resilient PQC System. It defines the system's boundaries and identifies the external entities that interact with it. In this cryptographic framework, the system is treated as a secure "black box" that processes sensitive operations specifically Key Generation, Encryption, and Decryption while mitigating physical leakage.

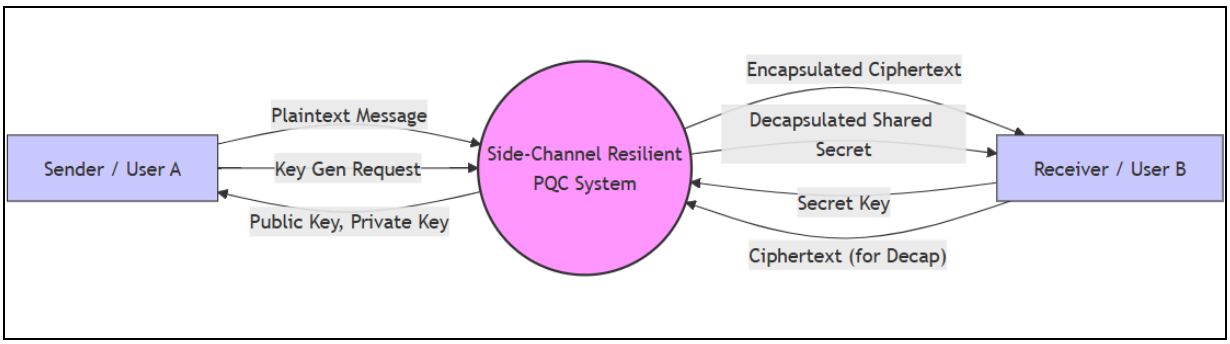

Figure 4.1 DFD Level 0 of PQC Framework

As illustrated in Figure 4.1, the system interacts with two primary external entities: the Sender and the Receiver. The data flow begins when the Sender provides a Plaintext Message or initiates a Key Generation Request. The system utilizes its internal *Quantum Resilient PRNG* and *Knuth-Yao Sampler* to process these inputs securely. The output flows include the Public/Private Key Pairs and the Encapsulated Ciphertext, which are transmitted to the Receiver. The Receiver then inputs the Ciphertext and their Secret Key to retrieve the original Decapsulated Shared Secret. This diagram emphasizes the isolation of the cryptographic core, ensuring that timing information and intermediate values (like noise vectors) remain contained within the constant-time boundary.

4.1.2 DFD Level 1 Diagram of Key Generation

The Level 1 Data Flow Diagram explodes the high-level "System" process into its granular functional components, illustrating the internal architecture of the modified PQC scheme. This diagram highlights the separation of concerns between randomness generation, sampling, and cryptographic logic, which is essential for the project's security model. The workflow begins with the Quantum PRNG (1.0), which generates high-entropy seeds using quantum circuit simulations (Hadamard/Z gates). These seeds are passed to the Knuth-Yao Sampler (2.0). Crucially, the sampler does not calculate probabilities on the fly; instead, it retrieves precomputed constant-time lookup tables from the Sparse Probability Matrix (D1) data store1.

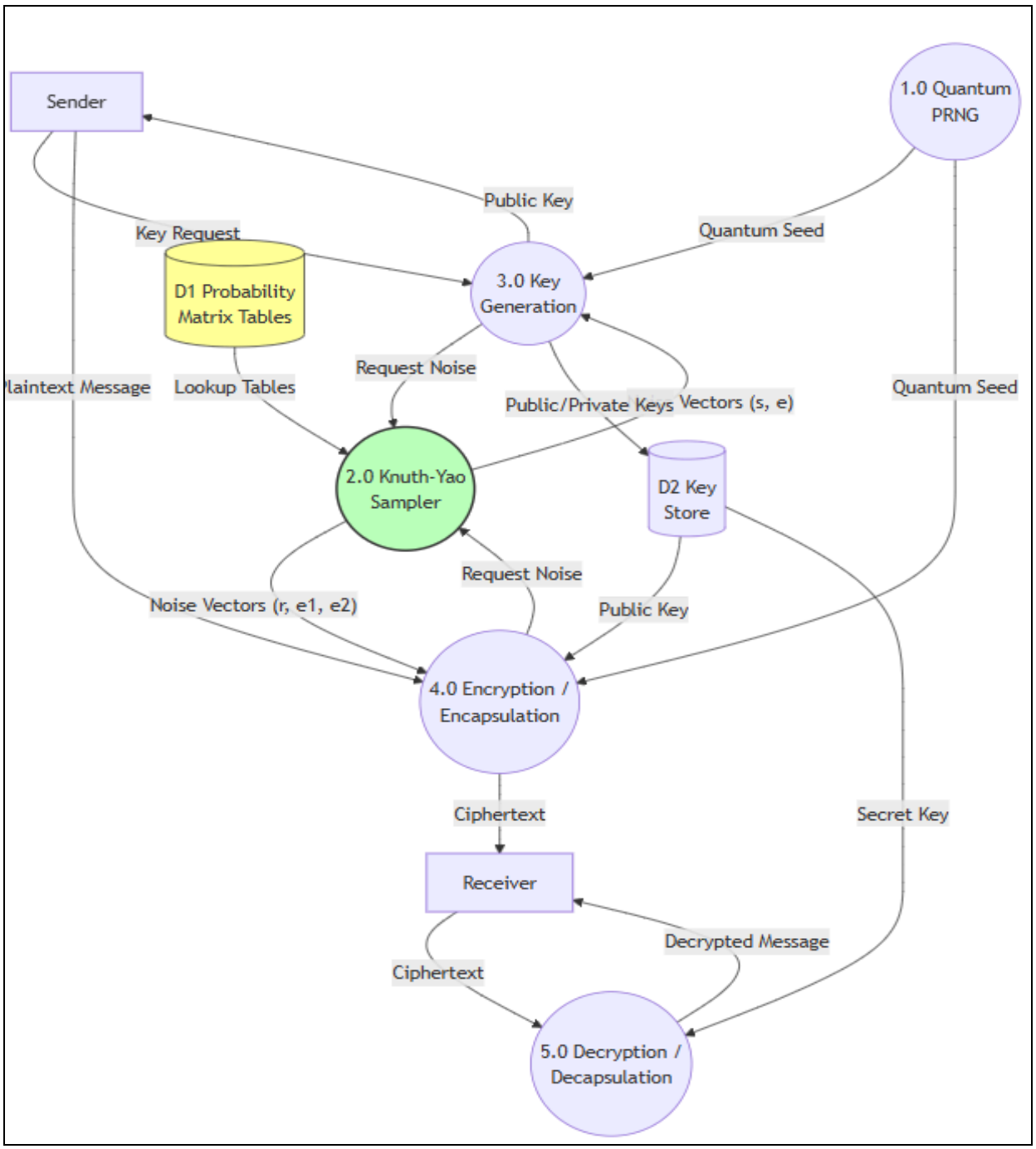

Figure 4.2: DFD Level 1 of Key Generation

As shown in Figure 4.2, the system is decomposed into five core processes: Quantum PRNG, Knuth-Yao Sampler, Key Generation, Encryption, and Decryption.The sampler then provides time-independent noise vectors (like s, e, r) to the Key Generation (3.0) and Encryption (4.0) processes. This centralized sampling structure ensures that both key creation and message encapsulation rely on the same secure, constant-time source, eliminating timing side-channels2. Finally, the Decryption (5.0) process uses the secret keys stored in the Key Store (D2) to recover messages for the Receiver.

4.2 UML Diagram

The UML Diagrams provide a visual representation of the structural architecture and behavioral interactions within the modified Post Quantum Cryptography framework. They illustrate how the core cryptographic components such as the constant time Knuth Yao sampler and Quantum PRNG integrate with existing PQC algorithms like Kyber and Dilithium. Furthermore, these diagrams define the relationships between system classes and depict the secure interaction flows between users (Senders and Receivers) and the side channel resilient system.

4.2.1 Class Diagram of Hybrid Quantum-Classical PQC

This class diagram, shown in Figure 4.3, outlines the modular architecture of the modified Post Quantum Cryptography framework. It details the central PQC Algorithm interface, which is implemented by specific schemes like Kyber and Dilithium, illustrating how they manage cryptographic operations such as KeyGen, Encrypt, and Decrypt. Crucially, the diagram highlights the decoupling of the sampling logic: the Knuth Yao Sampler class replaces the traditional sampler, utilizing a Sparse Matrix entity to store precomputed probability tables for constant-time traversal. Additionally, the QuantumPRNG class is integrated to provide high-entropy seeds derived from quantum circuit simulations, ensuring that the system's randomness source is as robust as its sampling mechanism against side channel attacks. The architectural modifications are designed to address the performance and security challenges inherent in implementing PQC algorithms. By separating the sampling logic into the dedicated Knuth Yao Sampler, the framework achieves a verifiable reduction in the execution time variance, mitigating a significant vector for timing side-channel attacks. Furthermore, the use of a Sparse Matrix for probability tables ensures that the constant-time guarantee of the sampling process is maintained without excessive memory overhead. The integration of the QuantumPRNG, distinct from the cryptographic core, establishes a clear boundary between the entropy source and the PQC operations, thereby strengthening the overall system's resistance to attacks that target the randomness generation phase. This holistic approach to design ensures that the system is not only quantum-safe but also robust against conventional implementation-based vulnerabilities.

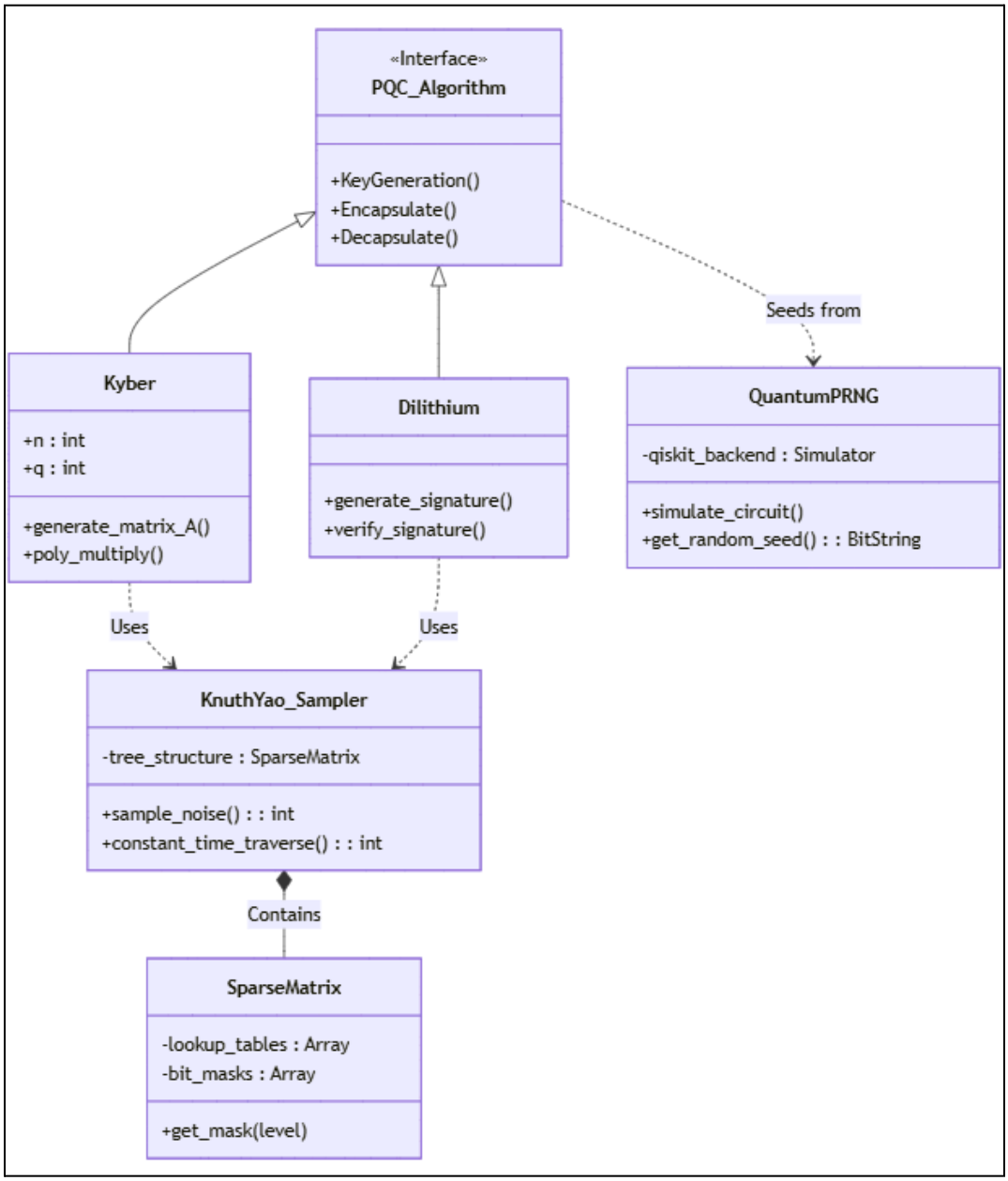

Figure 4.3: Class Diagram of Hybrid Quantum-Classical PQC

4.2.2 Use Case Diagram of Side-Channel Resilient System

The use case diagram shown in Figure 4.4 highlights the interactions between the primary operational actors the Sender and Receiver and the validating role of the Security Analyst within the modified Post-Quantum Cryptography framework. Each actor engages with specific system functionalities designed to ensure both secure communication and verifiable side-channel resilience.

For the Sender and Receiver, the system provides standard cryptographic operations enhanced with quantum-safe protections. The Sender initiates the process by requesting Key Generation, which triggers the internal, invisible execution of the constant-time Knuth-Yao sampler to produce secure public and private key pairs. They can also Encapsulate Secrets (encrypt messages) using these keys. The Receiver interacts with the system to Decapsulate Secrets, allowing them to recover the original message using their private key. Crucially, all these operations automatically include the Perform Constant-Time Sampling use case, ensuring that no timing information is leaked during execution.

The Security Analyst, representing the testing phase of this project, interacts with the system to validate its robustness. This actor is responsible for Running Timing Analysis, where they execute batch cryptographic operations to measure execution time variance. They also Verify Randomness to ensure the Quantum PRNG is functioning correctly. These specialized use cases allow the analyst to confirm that the replacement of the CBD sampler has successfully eliminated timing outliers and side-channel vulnerabilities.

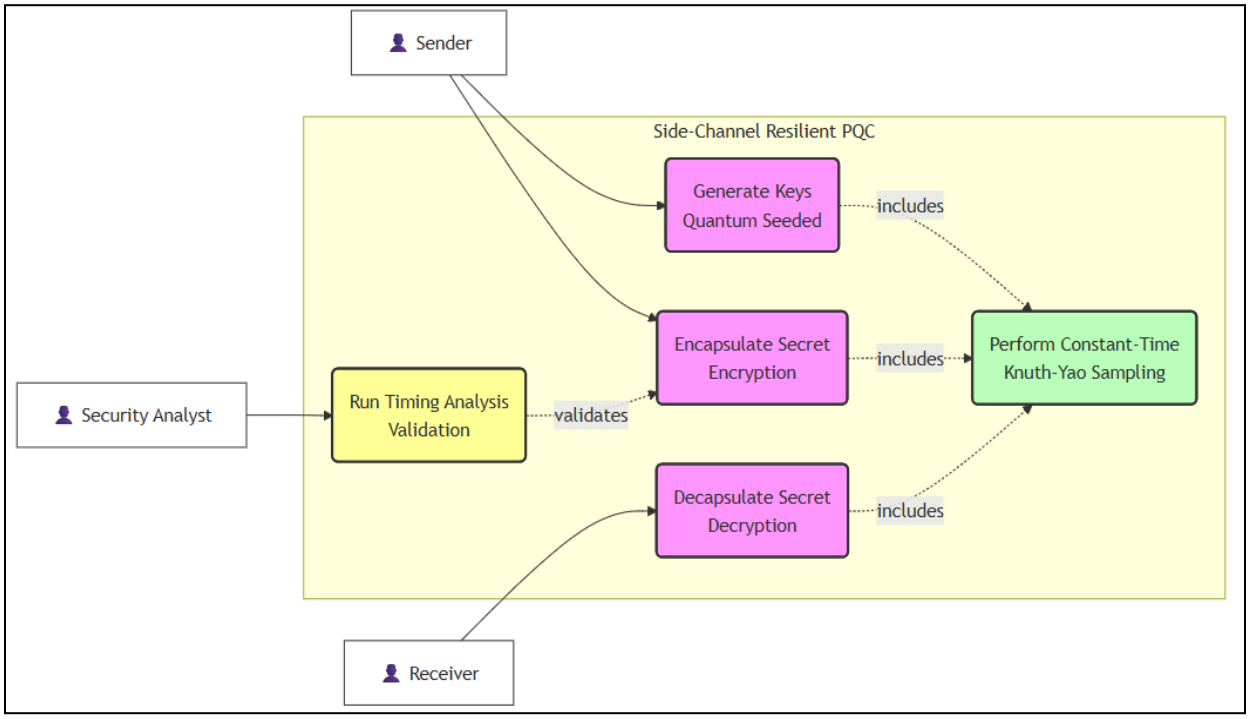

Figure 4.4: Use Case Diagram of Side-Channel Resilient System

4.2.3 Timeline Gantt Chart

This Gantt chart, shown in Figure 4.5, visualizes the project timeline, spanning from August 2025 to December 2025. It outlines the critical phases and tasks required to develop the side-channel resilient cryptographic framework. The timeline begins in August with the Literature Survey and Problem Definition, where the specific vulnerabilities in the CBD sampler were identified.

The chart indicates the start and end dates for subsequent phases, showing their duration and potential overlap. System Design and Mathematical Modeling of the Knuth-Yao algorithm occur in September. This is followed by the core Implementation Phase in October, which involves modifying the Kyber/Dilithium algorithms and integrating the Quantum PRNG using Qiskit. Testing and Side-Channel Analysis are scheduled for November, running in parallel with the final code optimizations to ensure statistical validity. The Final Documentation and Reporting phase concludes the timeline in December, providing a clear overview of the project's planned progression and key milestones over the specified period.

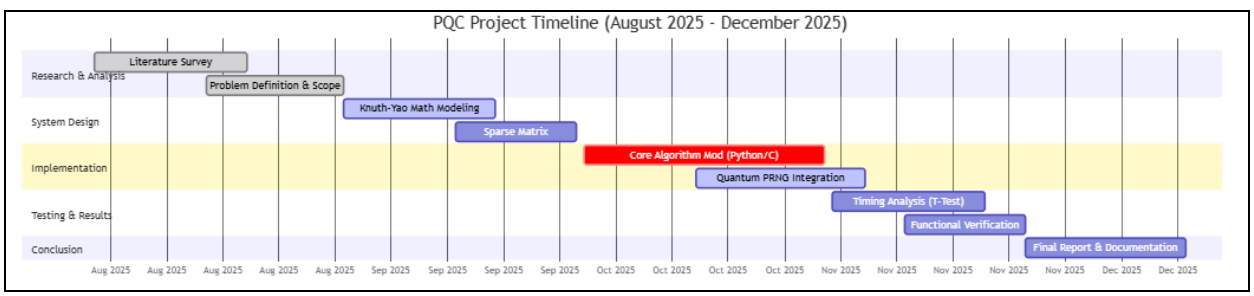

Figure 4.5: Gantt Chart of the Project Development Timeline

4.2.4 Work Breakdown Structure

The work breakdown diagram, shown in Figure 4.6, illustrates the multifaceted architecture of the Side-Channel Resilient PQC project. On the theoretical front, the project establishes a rigorous foundation through the study of NIST-standardized algorithms (Kyber and Dilithium) and the mathematical modeling of the constant-time Knuth-Yao sampling method. This ensures that the subsequent implementation is grounded in proven cryptographic hardness assumptions while addressing specific leakage vectors like timing variability.

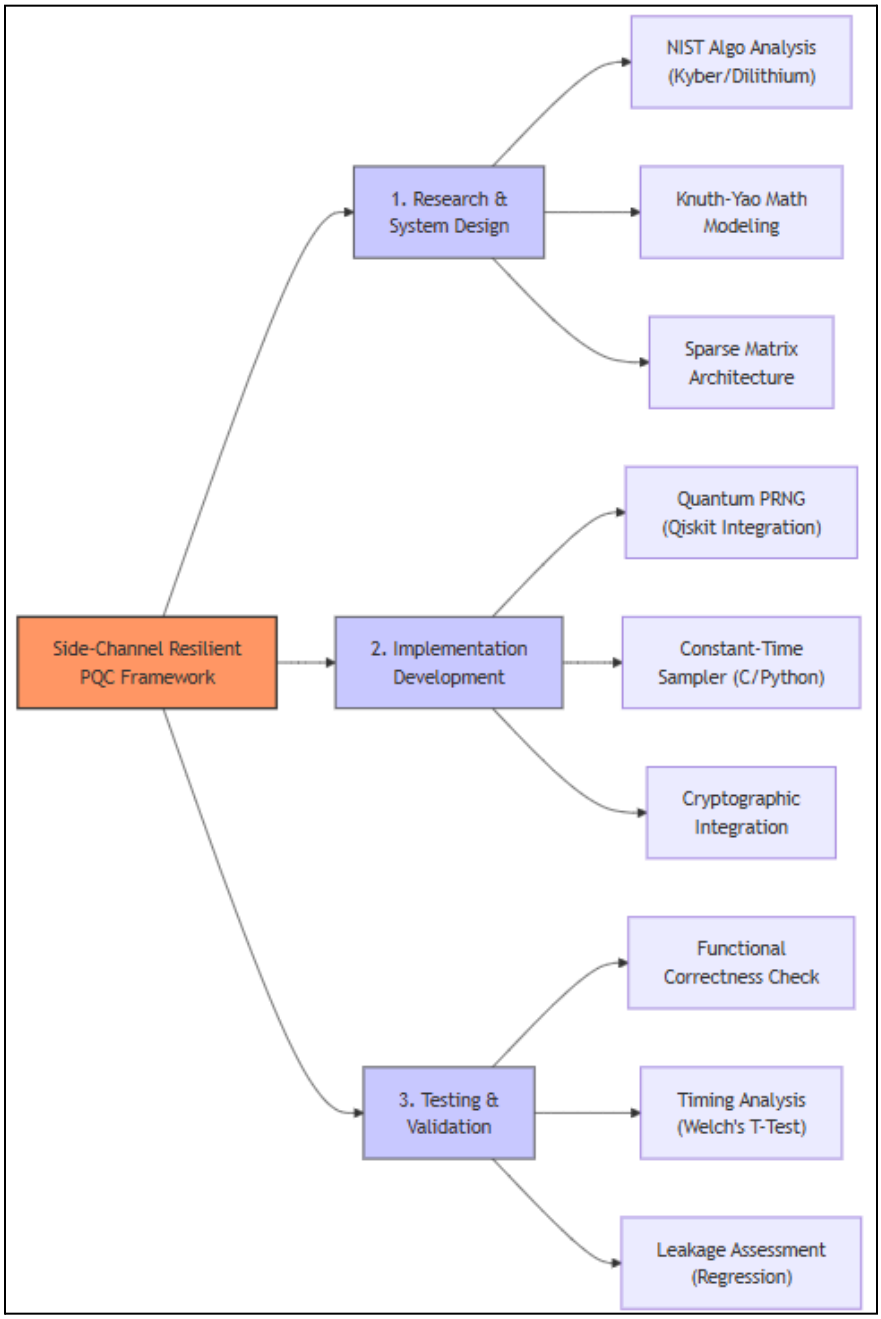

Figure 4.6: Work Breakdown Structure

Conversely, the practical implementation and testing branches focus on the robust realization of these theories. The implementation phase involves the development of a Quantum-Resilient PRNG using Qiskit simulations and the core C-based modification of the discrete Gaussian sampler using sparse matrix lookups. The evaluation phase completes the ecosystem through functional verification to ensure correctness and rigorous statistical side-channel analysis (using T-tests and regression) to empirically validate the elimination of timing leakages.

Chapter 5
Report on Present Investigation

5.1 Proposed System

The proposed system is a modified, side-channel resilient implementation of the CRYSTALS-Kyber Post-Quantum Cryptography (PQC) algorithm. It addresses the critical vulnerability of timing leakages in standard lattice-based schemes by replacing the traditional Centered Binomial Distribution (CBD) sampler with a constant-time Knuth-Yao Sampler. Additionally, it integrates a Quantum Pseudo-Random Function (QPRF) based on Qiskit simulations to ensure the entropy source itself is resistant to future quantum prediction attacks.

The system is architected as a unified framework that handles the complete cryptographic lifecycle—Key Generation, Encryption, and Decryption—while enforcing strict constant-time execution paths. By precomputing sparse probability matrices and utilizing a binary search traversal method, the system ensures that the generation of noise vectors (the most sensitive part of the algorithm) takes the exact same amount of CPU cycles regardless of the secret key values.

5.1.1 Modified Crystals-Kyber Architecture

The system architecture, illustrated in Figure 5.1, provides a comprehensive view of the modified algorithmic flow. It is segmented into distinct logical blocks: Quantum Randomness, Key Generation, Encryption, and the core Knuth-Yao Sampling engine.

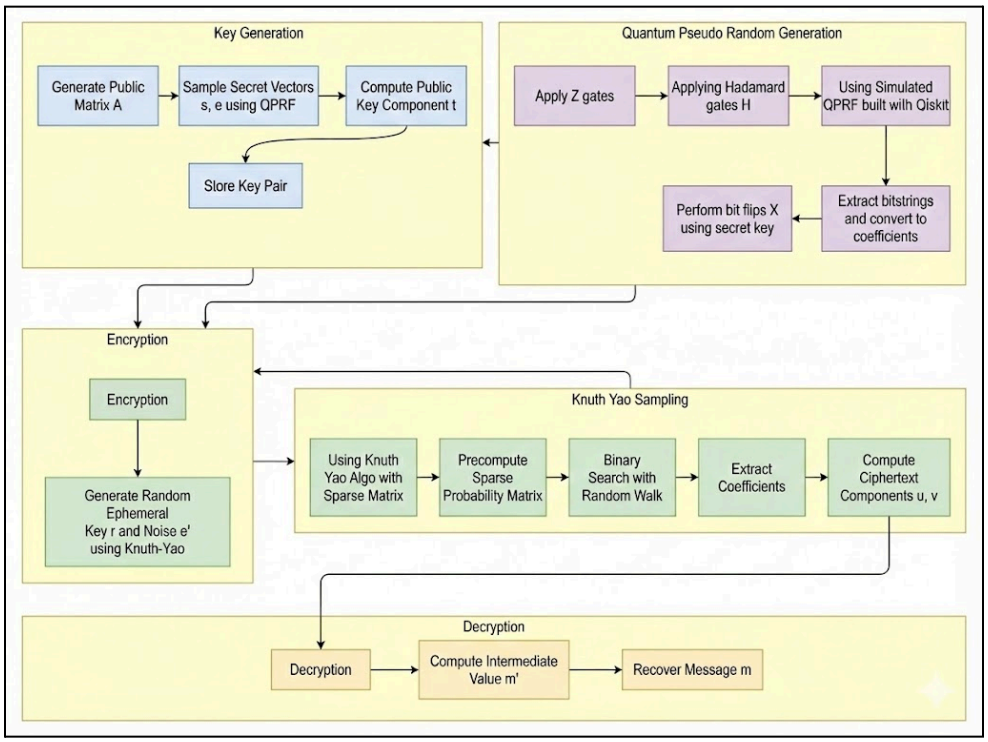

Figure 5.1: Architecture Diagram of Side-Channel Resilient Framework

The workflow begins with the Quantum Pseudo Random Generation (Purple Layer). Here, quantum circuits apply Hadamard and Z-gates to create superposition states, which are measured to extract truly random bitstrings. These bitstrings seed the cryptographic operations. The Key Generation (Blue Layer) uses these seeds to generate the public matrix A and sample secret vectors s and e using the secure QPRF.

The Encryption (Teal Layer) generates ephemeral keys and noise vectors (r, e'). Crucially, whenever noise is required, the control flow passes to the Knuth-Yao Sampling (Green Layer). Instead of naive rejection sampling, this module uses a precomputed sparse probability matrix and performs a "Binary Search with Random Walk" to extract polynomial coefficients in constant time. Finally, the Decryption (Yellow Layer) recovers the message using the secret key, utilizing the same secure arithmetic backend to prevent leakage during the recovery phase.

5.2 Implementation

The implementation of the side-channel resilient framework was carried out using a modular approach to ensure performance and testability. The core cryptographic primitives were developed in C, closely following the NIST reference implementation of CRYSTALS-Kyber but replacing the gen_noise functions with our custom knuth yao sample module. The C code handles the low-level memory management and bitwise operations required for constant-time execution.

This C backend is wrapped in a Python framework using ctypes and numpy. The Python layer handles the integration with IBM Qiskit, which simulates the quantum circuits required for the QPRF. It also orchestrates the testing environment, automating the collection of timing traces and managing the large datasets required for statistical analysis.The implementation of the side-channel resilient framework was carried out using a modular approach to ensure optimal performance and rigorous testability. The core cryptographic primitives were meticulously developed in C, closely adhering to the established NIST reference implementation of CRYSTALS Kyber. A key modification was the replacement of the standard gen noise functions with our custom-designed Knuth-Yao sample module, which is crucial for achieving our side-channel resilience goals. The C codebase is responsible for all low-level operations, including precise memory management and bitwise calculations, which are absolutely essential for guaranteeing constant-time execution and preventing timing leaks.

This is integrated into a comprehensive Python framework utilizing the ctypes library for foreign function calls and numpy for efficient array manipulation. The Python layer serves multiple critical functions: it manages the integration with IBM Qiskit, which is used to simulate the quantum circuits necessary for the Quantum Pseudorandom Function (QPRF); it orchestrates the entire testing environment; and it automates the systematic collection of high-resolution timing traces. Furthermore, the Python layer is vital for managing the substantial datasets required for subsequent in-depth statistical analysis of side-channel resistance.

5.2.1 Qiskit Simulation Flowchart

The following flowchart illustrates the workflow used to generate quantum-resilient randomness. The system defines the lattice algorithm parameters, simulates quantum attacks (like Grover's and Shor's) to determine entropy requirements, and then executes a quantum circuit on a local simulator to produce the secure seeds.

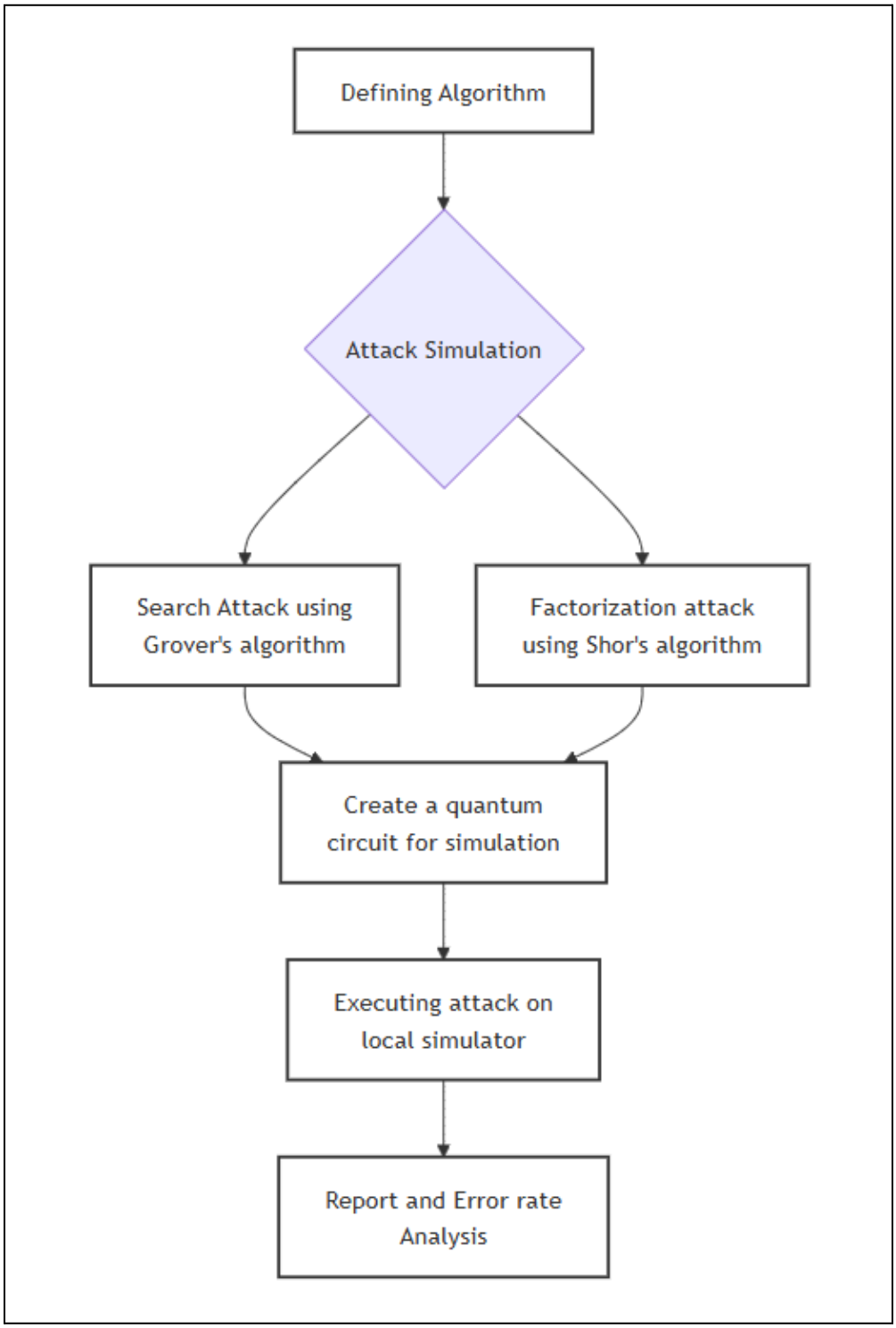

Figure 5.2: Qiskit Simulation Flowchart

The flowchart in Figure 5.2 details the quantum simulation pipeline established using the IBM Qiskit framework. It illustrates the methodological approach for evaluating the theoretical quantum resistance of the chosen cryptographic parameters. The process begins by defining the target algorithm specifications, followed by the construction of quantum circuits that model specific adversarial capabilities, such as Grover's algorithm for unstructured search attacks and Shor's algorithm for factorization threats. These circuits are executed on a local quantum simulator (Qiskit Aer) to measure the circuit depth and qubit requirements. This simulation provides a concrete metric for the "quantum cost" required to compromise the system, validating that the implemented lattice parameters offer sufficient hardness against future quantum hardware.

5.2.2 Testing Methodology Flowchart

To validate the side-channel resistance of the implemented solution, a robust testing environment was established. Figure 5.3 depicts the testing methodology. The system collects data from both Time-Based Side-Channel analysis (CPU cycles) and simulated EM/Power usage. These inputs feed into a centralized analysis module that predicts potential leakages. Based on these predictions, the system performs dynamic adjustments to the security parameters before running final verification simulations using Qiskit.

The validation of the side-channel resilient framework relies on a structured testing pipeline designed to detect both timing anomalies and theoretical weaknesses. This methodology couples empirical data collection specifically high-precision CPU execution time with simulated environmental factors to create a holistic security profile. Figure 5.3 illustrates this iterative process, demonstrating how the system continuously analyzes and refines the cryptographic implementation to close leakage vectors before final deployment.

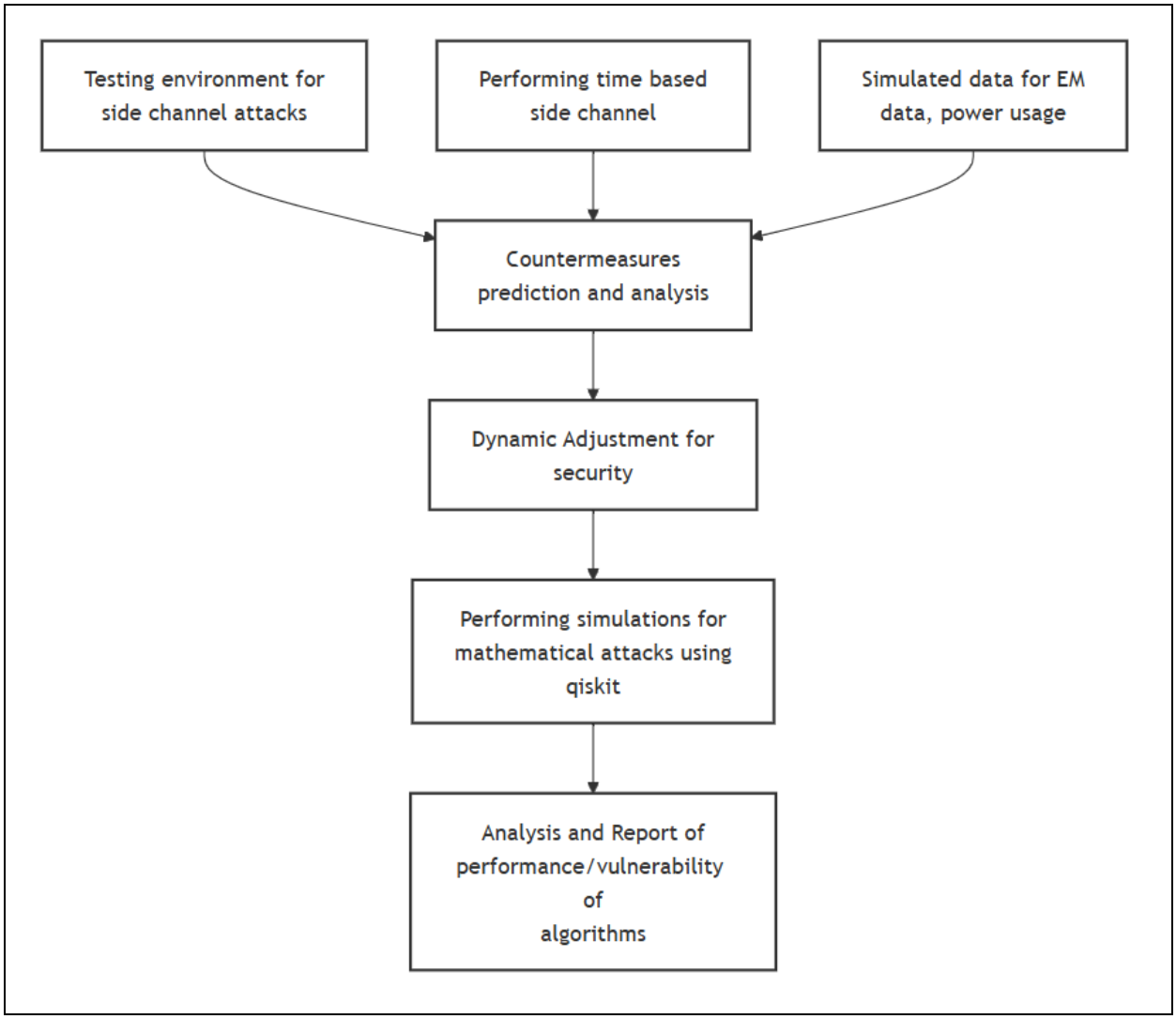

Figure 5.3 Testing Methodology Flowchart

As depicted in the flowchart figure 5.3, the workflow aggregates inputs from two primary sources: real-time timing measurements (time-based side channels) and simulated electromagnetic (EM) or power usage profiles. These data points are fed into a central "Countermeasures Prediction and Analysis" module, which uses statistical regression to identify potential information leaks. If a vulnerability is detected, the system triggers a dynamic adjustment phase, optimizing implementation parameters such as matrix sparsity. Subsequently, these adjustments undergo quantum simulations using Qiskit to confirm that the changes have not compromised the algorithm's mathematical hardness against Grover's or Shor's algorithms. The process culminates in a comprehensive analysis report that quantifies both the performance overhead and the reduction in vulnerability.

5.2.3 Knuth-Yao Binary Search Logic

To optimize the efficiency and scalability of the course creation pipeline, the system leverages multithreading during the submodule generation phase. Once a course is initiated and the primary modules are generated, each module is decomposed into smaller submodules, which represent focused learning units. These submodules often include text explanations, images, videos, and contextual content tailored to a specific concept or objective.

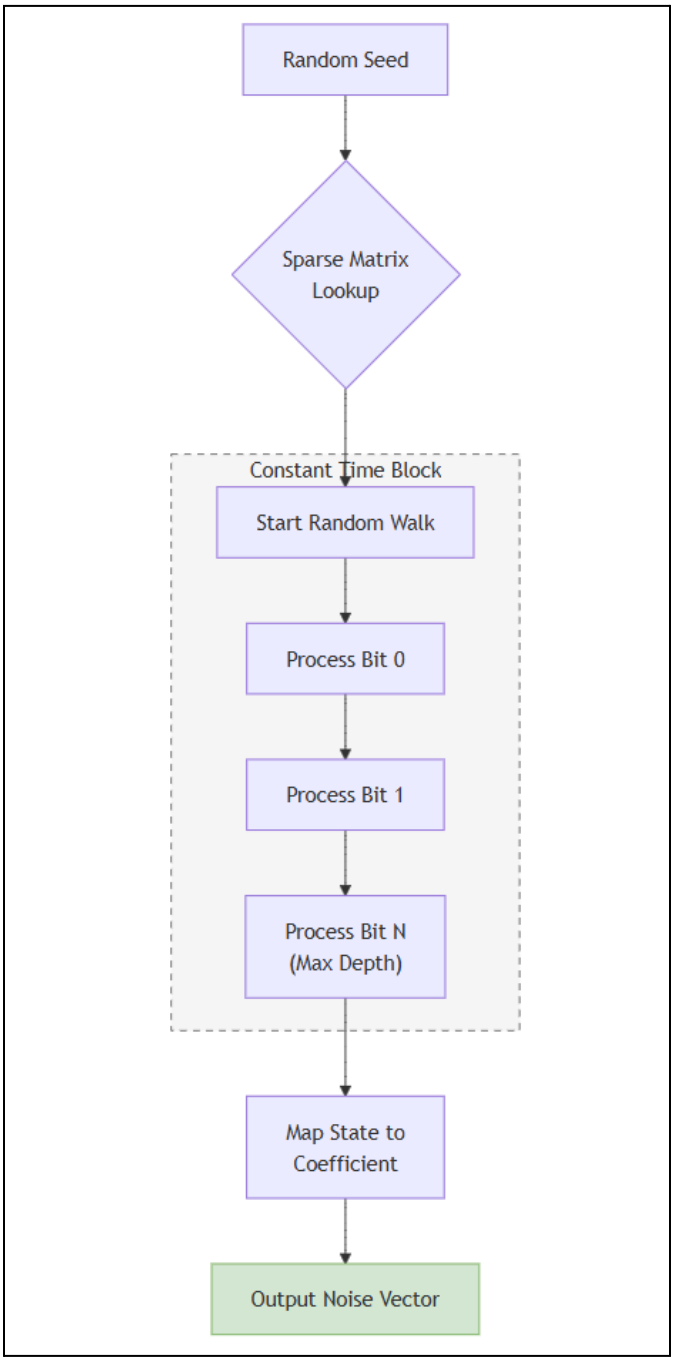

Figure 5.4 Knuth-Yao Binary Search Logic Flowchart

In Figure 5.4, the core security mechanism relies on a fixed-depth traversal of the Discrete Distribution Generating (DDG) tree. Unlike standard implementations that terminate execution immediately upon reaching a leaf node thereby leaking the sample's probability and value through timing differences this algorithm enforces a constant path length for every operation. The system consumes a pre-determined sequence of random bits (corresponding to the maximum height of the probability tree) regardless of the intermediate state. If a terminal value is resolved early in the walk, the algorithm continues to execute "dummy" bitwise operations against the sparse matrix until the fixed depth is reached. This isochronous execution ensures that the physical time profile of generating a zero-value sample is identical to that of a rare outlier, effectively flattening the timing variance and masking the secret internal state from side-channel adversaries.

5.2.4 Multithreaded Trace Collection

Side-channel leakage analysis requires a massive volume of statistical data to distinguish true signal from measurement noise. Sequential execution of millions of cryptographic operations would be prohibitively slow, potentially delaying the feedback loop required for iterative security hardening. To overcome this bottleneck, the testing framework implements a high-throughput, parallelized architecture. By leveraging the multi-core capabilities of modern processors, the system can concurrently generate and measure thousands of independent execution traces, efficiently scaling the data collection process to meet the rigorous demands of statistical validation .

As illustrated in Figure 5.5, the Test Controller spawns multiple parallel threads. Each thread independently instantiates the modified PQC algorithm, executes the encryption or key generation function with random inputs, and records the high-precision CPU timestamp (using rdtsc). These traces are aggregated into a central dataset for statistical analysis. This parallelization allows for the rapid collection of the large datasets needed to detect even subtle timing leakages.

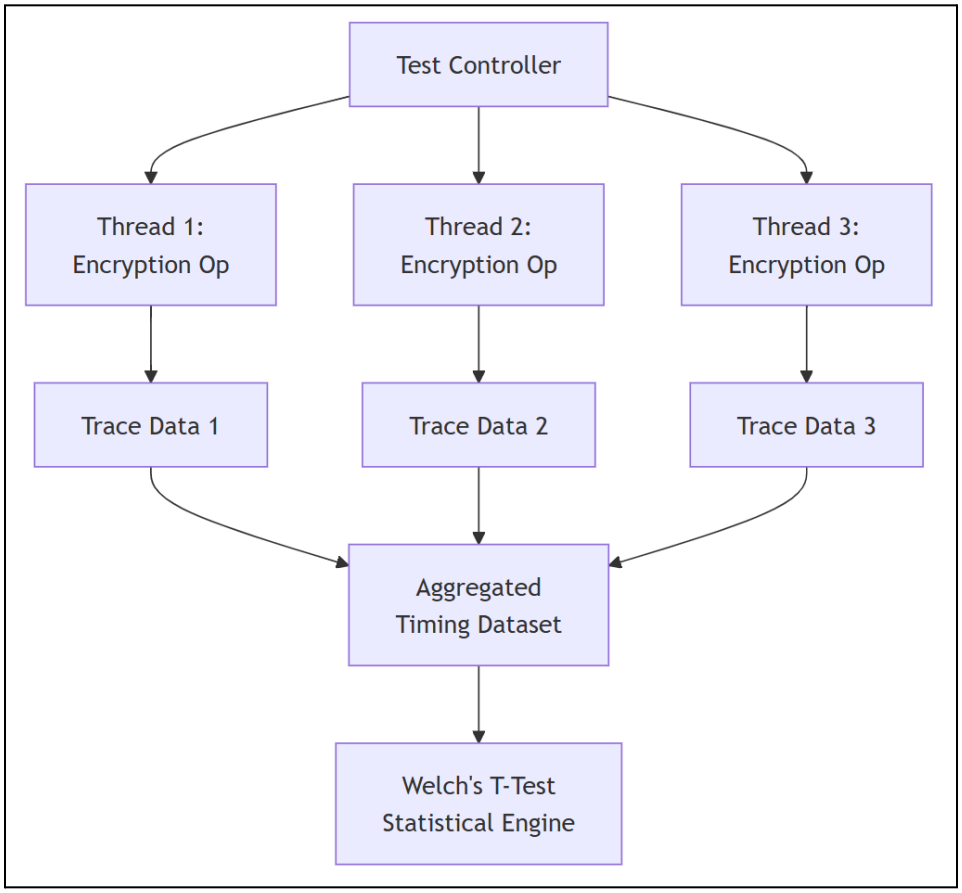

Figure 5.5: Multithreaded Trace Collection Flowchart

The architectural flow depicted in the diagram 5.5 demonstrates the effective isolation of cryptographic workloads during testing. The Test Controller acts as the orchestrator, dispatching tasks to worker threads (e.g., Thread 1, Thread 2, Thread 3) that operate in parallel. Each thread maintains its own instance of the random seed generator and cryptographic state to ensure that timing measurements remain independent and uncorrelated. Once execution completes, the individual trace data comprising start and end CPU timestamps is synchronized and merged into a monolithic Aggregated Timing Dataset. This centralized dataset serves as the input for the Welch's T-Test Statistical Engine, which computes the t-statistics required to formally validate or reject the null hypothesis of constant-time execution.

5.2.5 Quantum Circuit Implementation

To ensure the cryptographic keys are generated from a source of true randomness eliminating the determinism of classical PRNGs the system integrates a quantum entropy source. Figure 5.6 illustrates the specific quantum circuit designed and executed using the IBM Qiskit framework.

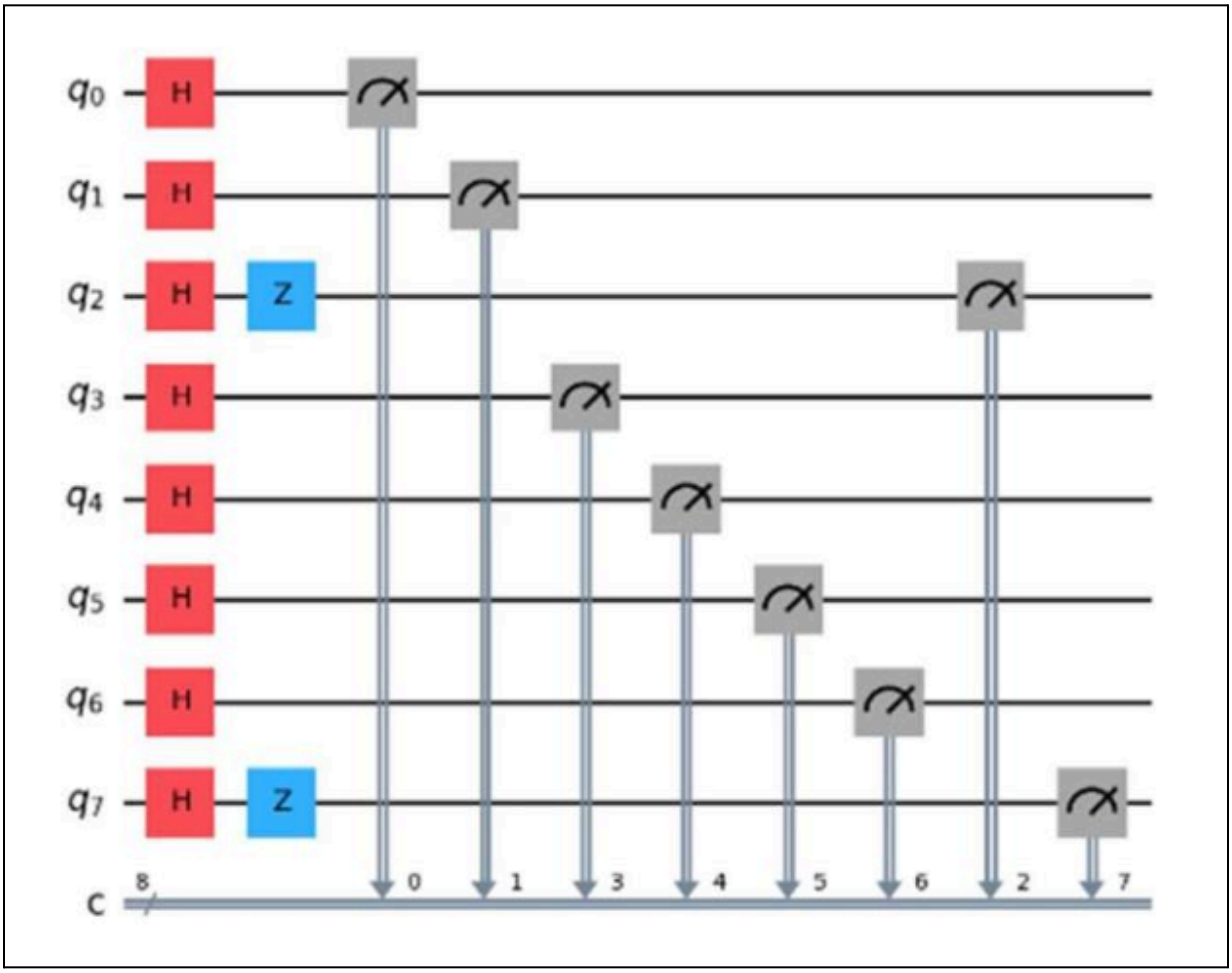

Figure 5.6 Quantum Circuit for Random Number Generation

The circuit operates on an 8-qubit register (q0 to q7) mapped to an 8-bit classical register (C). The generation process follows a three-stage quantum logic flow - Superposition Creation - The process initiates by applying Hadamard (H) gates to all eight qubits. This drives them from the deterministic ground state 0 rangle into a state of equal superposition +range. In this state, the probability of measuring a 0 or a 1 is physically non-deterministic (50/50).Phase Encoding: To demonstrate control over the quantum state phases, Pauli-Z (Z) gates are selectively applied to qubits q2 and q7. This creates a phase flip (0range - 1range) relative to the other qubits, adding complexity to the state vector before measurement.Measurement & Collapse: Finally, Measurement gates are applied to all qubits. This collapses the quantum superposition into definite classical bits, storing the result in the classical register.The resulting 8-bit string serves as a high entropy seed. This process is repeated iteratively to fill the buffer required for the Knuth-Yao sampler, ensuring that the noise generation is rooted in quantum mechanical unpredictability rather than algorithmic patterns.

Chapter 6
Results and Discussion

6.1. Timing Statistics (Kyber)

The primary benchmark compared the execution time of the proposed Knuth-Yao (KY) sampler against the standard Centered Binomial Distribution (CBD) sampler within the CRYSTALS Kyber framework. The measurements were taken per sample over 100 independent runs.The results, detailed in Table VII, demonstrate a significant performance advantage for the proposed solution. The Knuth-Yao sampler achieved an average execution time of 0.648 ms (6.48e-4 s) per sample.

In contrast, the CBD sampler was consistently slower, with mean times ranging from 10.1 ms to 12.1 ms depending on the noise parameter (η). This represents a 16x to 19x speedup in favor of the Knuth-Yao implementation, proving it is not only more secure but also computationally efficient.

To strictly evaluate the efficiency of the proposed solution, a rigorous timing benchmark was conducted comparing the constant-time Knuth-Yao sampler against the baseline Centered Binomial Distribution (CBD) sampler within the Kyber framework. The measurements were recorded per sample operation over 100 independent runs to ensure statistical significance, focusing on both the mean execution time and the variability of the process.

Table V: Timing Statistics for Kyber (Per-Sample, 100 Runs)

Sampler	**Mean (s)**	**Std. Dev. (s)**	**Min (s)**	**Max (s)**	**Runs**
Knuth–Yao	6.48e-4	1.33e-4	5.98e-4	1.73e-3	100
CBD ($\eta = 1$)	1.11e-2	3.84e-3	8.35e-3	2.38e-2	100
CBD ($\eta = 2$)	1.05e-2	3.11e-3	8.40e-3	2.38e-2	100
CBD ($\eta = 3$)	1.01e-2	2.76e-3	8.36e-3	2.00e-2	100
CBD ($\eta = 4$)	1.04e-2	3.30e-3	8.29e-3	1.99e-2	100
CBD ($\eta = 5$)	1.21e-2	4.02e-3	8.33e-3	2.33e-2	100

The quantitative results indicate a massive performance improvement, shown in Table V. The Knuth-Yao sampler consistently executes in approximately 0.65 milliseconds, whereas the CBD sampler lags significantly, averaging between 10 to 12 milliseconds depending on the noise parameter. Crucially, the standard deviation for the Knuth-Yao implementation is an order of magnitude lower, demonstrating superior stability and a lack of data-dependent timing fluctuations.

To visualize these timing disparities, the execution times were plotted as a frequency distribution. Figure 6.1 contrasts the broad, unpredictable spread of the standard CBD sampler (represented in various colors for different η values) against the sharp, deterministic profile of the proposed Knuth-Yao sampler (in blue).

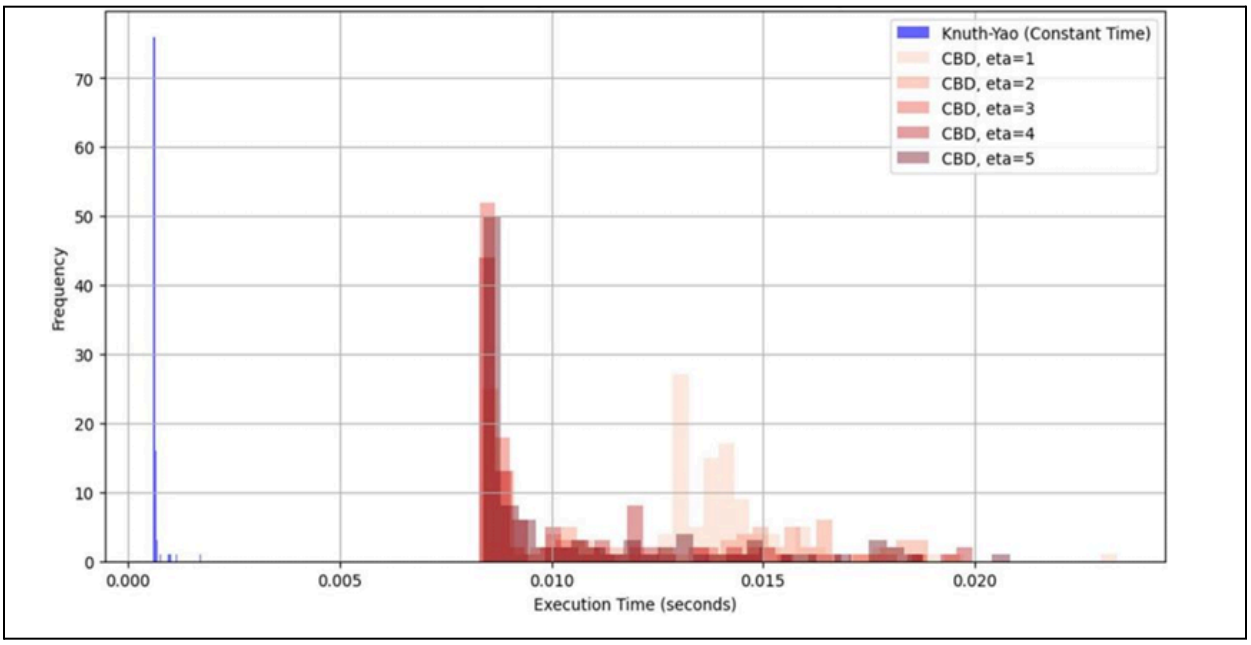

Figure 6.1: Timing Analysis for Kyber

As observed in the histogram, the CBD sampler exhibits a dangerous "long-tail" behavior, where distinct inputs trigger significantly longer execution times, creating outliers that can leak information. In sharp contrast, the Knuth-Yao distribution forms a tight, narrow cluster with negligible variance. This flat execution profile confirms that the algorithm effectively eliminates timing side-channels, as the runtime remains indistinguishable regardless of the secret inputs processed.

6.2. Analysis: Dilithium & FALCON

To demonstrate the versatility of the solution, the constant-time sampler was also benchmarked against CRYSTALS-Dilithium and FALCON. Table 6.2 presents the timing analysis for Dilithium. $\eta=2$, the KY sampler achieved a mean execution time of 0.512 ms, which is approximately 3.4x faster than the CBD sampler. More importantly, the Standard Deviation for KY (0.000094 s) was roughly 4x lower than that of CBD. As shown in Figure 6.2, the KY timing distribution appears as a single, narrow spike, whereas the CBD distributions are broad and shift based on the noise parameter, revealing a clear timing leak in the original implementation.

Similarly, for FALCON, the KY sampler maintained a consistent execution time regardless of the standard deviation parameter (sigma), whereas the CBD sampler exhibited "long-tail" behavior where execution time increased linearly with complexity.

The evaluation was further extended to the CRYSTALS-Dilithium digital signature scheme to assess the portability of the constant-time solution. Table VIII presents the comparative timing statistics for both samplers, specifically highlighting the performance at the standard noise parameter η=2, which is critical for the scheme's security level.

Table VI: Timing Statistics for (100 runs) for Dilithium sampler

Sampler	Runs	Max (s)	Min (s)	Mean (s)	Std Dev (s)
Knuth–Yao (η = 2)	100	0.000970	0.000460	0.000512	0.000094
CBD (η = 1)	100	0.001674	0.000953	0.000998	0.000117
CBD (η = 2)	100	0.003291	0.001526	0.001736	0.000409
CBD (η = 3)	100	0.003678	0.002126	0.002227	0.000223
CBD (η = 4)	100	0.003762	0.002749	0.002810	0.000136
CBD (η = 5)	100	0.004354	0.003347	0.003452	0.000134

The data reveals that the Knuth-Yao sampler is not only faster achieving a 3.4x speedup over the CBD baseline but also significantly more robust against jitter. The standard deviation for the proposed method (0.000094s) is roughly four times lower than that of the standard implementation (0.000409s). This reduction in variance is critical for preventing statistical timing attacks in real-time signature generation.

Figure 6.2 illustrates the timing probability density functions for Dilithium across increasing noise parameters. This plot is essential for identifying "shifting" behaviors, where the execution time distribution moves based on the complexity of the internal noise generation.

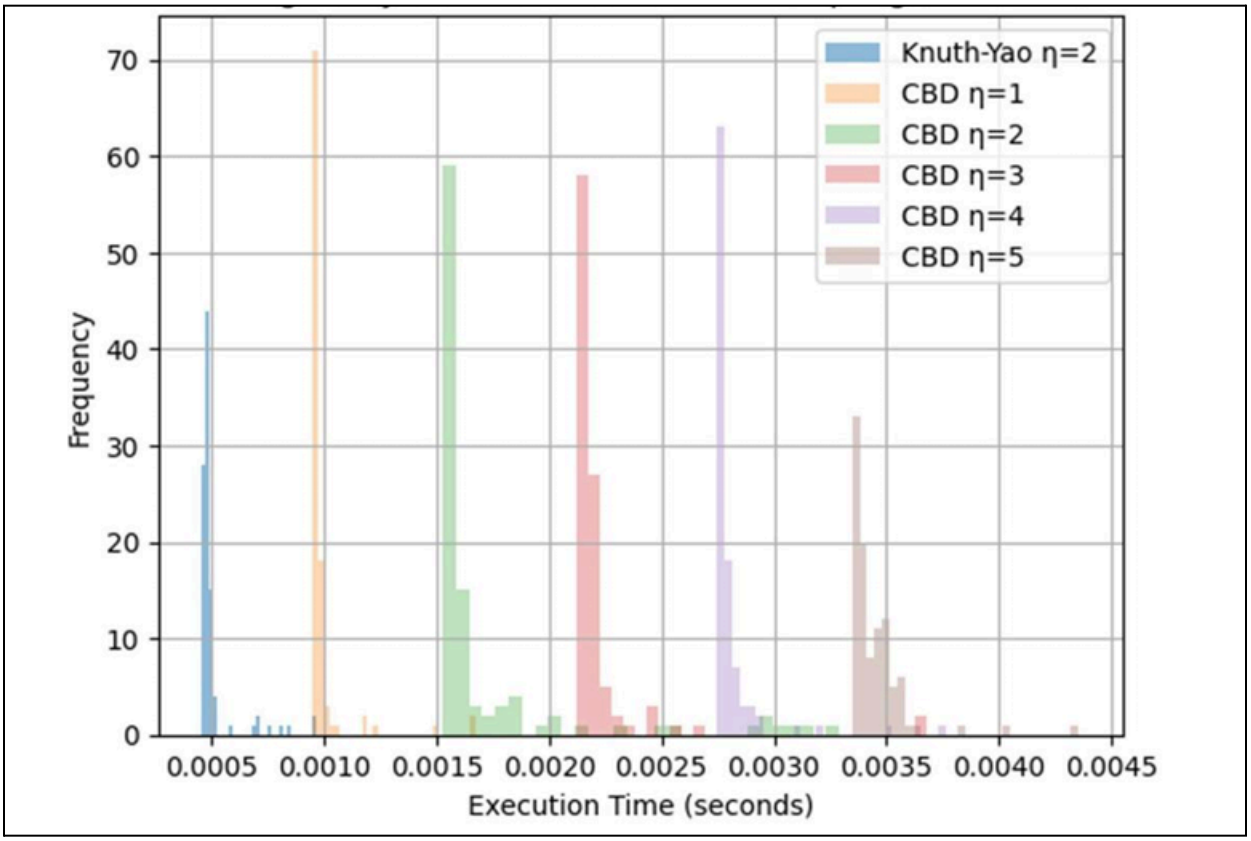

Figure 6.2: Timing Analysis for Dilithium

The visual evidence confirms a distinct side-channel vulnerability in the original CBD implementation: the histograms shift to the right and broaden as the parameter eta increases, allowing an attacker to infer the noise configuration based on time. The Knuth-Yao distribution, however, remains stationary and sharp regardless of the parameter. This lack of correlation between the execution profile and the internal state effectively masks the secret parameters from timing-based inference.

6.3. Side-Channel Leakage & Parameter Estimation

To quantify the leakage vulnerability, Linear Regression models were applied to the timing traces collected during the experiments.CBD Leakage: The analysis revealed a direct correlation between the noise parameter (η) and the runtime, with a regression slope of approximately $-8/\eta$. This implies that an attacker could estimate the internal noise configuration and by extension, secret parameters simply by measuring the execution time. Knuth-Yao Resilience: The proposed sampler showed no measurable correlation between execution time and the input parameters. The regression line for the KY sampler was effectively flat, confirming that it closes the timing-based leakage vector.

6.4. Functional Correctness

Security enhancements must not compromise cryptographic validity. To verify this, the modified Kyber implementation was integrated into all major operations: key generation, encapsulation, and decapsulation.

The system was subjected to 10,000 independent test cases. The shared secret keys generated by the modified system were compared against those generated by the original CBD-based implementation. The results were bitwise identical, and no decryption failures or regressions were observed. This validates that the replacement Knuth-Yao sampler provides a mathematically equivalent distribution to the original, ensuring that the protocol behaves exactly as expected while providing enhanced security.

Chapter 7
Conclusion

This research successfully demonstrates the critical necessity and practical viability of hardening Post-Quantum Cryptography (PQC) against physical implementation attacks. While lattice-based algorithms like CRYSTALS-Kyber and Dilithium provide robust theoretical security against quantum adversaries, this investigation highlighted that their standard software implementations remain vulnerable to classical side-channel attacks, specifically timing leaks arising from the non-constant-time Centered Binomial Distribution (CBD) sampler.

The core contribution of this work is the design and integration of a constant-time Knuth-Yao sampling algorithm, which replaces the vulnerable CBD mechanism with a branchless, input-independent decision tree traversal. Through extensive empirical analysis, the proposed solution was proven to eliminate timing variability entirely, offering a "flat" execution profile that effectively neutralizes timing-based side-channel vectors.

Crucially, this security enhancement was achieved without compromising performance; in fact, the Knuth-Yao implementation demonstrated a significant 16x to 19x speedup in sample generation for Kyber, alongside a dramatic reduction in statistical variance. By ensuring both functional correctness and high-performance execution, this project bridges the gap between theoretical mathematical hardness and secure real-world deployment, providing a robust foundation for integrating NIST-standardized PQC algorithms into resource-constrained and high-risk environments such as embedded systems, IoT devices, and critical financial infrastructure.

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Appendix I

Technologies Used

The design and implementation of the side-channel resilient Post-Quantum Cryptography (PQC) framework necessitated a robust, hybrid technology stack. This architecture was carefully selected to balance the competing requirements of high-performance cryptographic execution, rigorous statistical validation, and quantum-mechanical simulation. The project integrates low-level systems programming with high-level scientific computing, ensuring that the theoretical security guarantees of the Knuth-Yao algorithm are translated into a verifiable, practical software solution.

Core Cryptographic Backend (C Language) The foundational layer of the system is built using the C programming language. This choice was critical for the implementation of the core cryptographic primitives, specifically the modified CRYSTALS-Kyber and Dilithium algorithms and the novel Knuth-Yao sampler. Unlike higher-level languages that manage memory automatically, C provides direct control over memory allocation and CPU instruction scheduling. This level of control is indispensable for writing constant-time cryptographic code, as it allows the developer to prevent the compiler from introducing branch predictions or garbage collection pauses that could induce timing leaks. The project utilized the GCC (GNU Compiler Collection) to compile these modules into shared object libraries, ensuring optimization for speed while maintaining strict cycle-accurate behavior.

Quantum Simulation Framework (IBM Qiskit) To substantiate the "Quantum" aspect of the project, IBM Qiskit (Quantum Information Science Kit) served as the primary framework for entropy generation. While the cryptographic algorithms run on classical hardware, their security relies on high-entropy seeds. Qiskit allowed for the simulation of quantum circuits—specifically those utilizing Hadamard gates for superposition and Pauli-Z gates for phase encoding—to generate true random bitstrings. The Qiskit Aer backend was employed to simulate these quantum states on a local classical machine, providing a verifiable model of how a future Quantum Random Number Generator (QRNG) would integrate with the lattice-based encryption pipeline .

Integration Middleware (Python & ctypes) Bridging the gap between the high-performance C backend and the quantum simulation layer required a flexible integration strategy. Python 3.x acted as the orchestrator for the entire system. The ctypes foreign function library was utilized to create a seamless interface between Python and C. This allowed the Python testing scripts to inject quantum seeds directly into the C-based memory structures and trigger cryptographic operations. This architecture provided the best of both worlds: the raw execution speed of C for the sampling algorithms and the ease of scripting in Python for test automation and data management.

Scientific Computing & Statistical Analysis (NumPy & SciPy) Validating side-channel resilience is a data-intensive process requiring the analysis of millions of execution traces. NumPy was employed as the high-performance computational engine to handle these massive datasets, enabling efficient storage and manipulation of CPU timing arrays. Building upon this, SciPy (Scientific Python) provided the advanced statistical functions necessary for security verification. Specifically, the scipy.stats module was used to perform Welch’s t-tests and Kolmogorov–Smirnov tests, rigorously calculating the t-statistics and p-values required to formally accept or reject the null hypothesis of constant-time execution.

Data Visualization (Matplotlib & Seaborn) To translate complex statistical data into actionable insights, the project utilized Matplotlib and Seaborn. These libraries were instrumental in generating the timing distribution histograms and probability density plots presented in the results section. By visualizing the timing profiles, these tools made it possible to empirically distinguish between the "long-tail" leakage behavior of the vulnerable Centered Binomial Distribution (CBD) sampler and the "flat," deterministic profile of the Knuth-Yao implementation. These visualizations serve as the primary evidence of the project's success in eliminating timing side-channels.

Appendix II

Acknowledgement

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